
Math 230B Notes

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Chapter 1

Differentiation

1.1 The Derivative of a Function

Definition 1.1.1. (Differentiability and the Derivative)

Let $I \subseteq \mathbb{R}$ be an interval, $f : I \rightarrow \mathbb{R}$, and $c \in I$.

(i) We say f is differentiable at c if

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$$

exists (that is, it equals a real number). In this case, the quantity $\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$ is called the derivative of f at c and is denoted by

$$f'(c), \quad \frac{df}{dx}(c), \quad \frac{df}{dx}|_{x=c}$$

(ii) If $f : I \rightarrow \mathbb{R}$ is differentiable at every point $c \in I$, we say f is differentiable (on I).

Remark 1.1.1. Note that

$$\begin{aligned} f'(c) = L &\iff \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = L \\ &\iff \forall \epsilon > 0 \exists \delta > 0 \text{ such that if } 0 < |x - c| < \delta, \text{ then } \left| \frac{f(x) - f(c)}{x - c} - L \right| < \epsilon \\ &\iff \forall \epsilon > 0 \exists \delta > 0 \text{ such that if } 0 < |h| < \delta, \text{ then } \left| \frac{f(c + h) - f(c)}{h} - L \right| < \epsilon \quad (\text{Let } h = x - c) \\ &\iff \lim_{h \rightarrow 0} \frac{f(c + h) - f(c)}{h} = L \end{aligned}$$

Remark 1.1.2. Let A denote the collection of all points at which $f : I \rightarrow \mathbb{R}$ is differentiable. If $A \neq \emptyset$, the function $f' : A \rightarrow \mathbb{R}$ defined by

$$f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \quad \forall c \in A$$

is called the derivative of f .

Example 1.1.1. Let $I \subseteq \mathbb{R}$ be an interval and $f : I \rightarrow \mathbb{R}$ be given by $f(x) = x^2$. Prove that f is differentiable on I and find the derivative.

Proof. $\forall c \in I$,

$$\begin{aligned} \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} &= \lim_{x \rightarrow c} \frac{x^2 - c^2}{x - c} \\ &= \lim_{x \rightarrow c} x + c \\ &= 2c \end{aligned} \quad (\text{ is continuous})$$

So, $\forall c \in I$ $f'(c) = 2c$. Hence,

$$f' : I \rightarrow \mathbb{R}, \quad f'(x) = 2x.$$

□

Example 1.1.2. Let $I \subseteq \mathbb{R}$ be an interval and $f : I \rightarrow \mathbb{R}$ be given by $f(x) = x^n$ where $n \in \mathbb{N}$, $n \geq 3$. Prove that f is differentiable on I and find the derivative.

Proof.

$$\begin{aligned} \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} &= \lim_{x \rightarrow c} \frac{x^n - c^n}{x - c} \\ &= \lim_{x \rightarrow c} \frac{(x - c)(x^{n-1} + cx^{n-2} + \dots + c^{n-1})}{x - c} && \text{(Algebra)} \\ &= \lim_{x \rightarrow c} [x^{n-1} + cx^{n-2} + \dots + c^{n-1}] \\ &= c^{n-1} + c \cdot c^{n-2} + \dots + c^{n-1} && \text{(Continuity)} \\ &= n \cdot c^{n-1} \end{aligned}$$

So, $\forall c \in I$ $f'(c) = n \cdot c^{n-1}$. Hence,

$$f' : I \rightarrow \mathbb{R}, \quad f'(x) = nx^{n-1}.$$

□

Example 1.1.3. Prove that $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = |x|$ is not differentiable at $c = 0$.

Proof. We need to show that $\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0}$ does not exist. Note that

$$\frac{f(x) - f(0)}{x - 0} = \frac{|x| - |0|}{x - 0} = \frac{|x|}{x}$$

Let $g(x) = \frac{|x|}{x}$. We want to show $\lim_{x \rightarrow 0} g(x)$ does not exist. By the sequential criterion for limits of functions, it is enough to find two sequences (a_n) and (b_n) in $\mathbb{R} \setminus \{0\}$ such that $a_n \rightarrow 0$ and $b_n \rightarrow 0$, but $\lim g(a_n) \neq \lim g(b_n)$. Let $a_n = -\frac{1}{n}$ and $b_n = \frac{1}{n}$. Clearly, $a_n \rightarrow 0$ and $b_n \rightarrow 0$. However,

$$\begin{aligned} \lim_{n \rightarrow \infty} g(a_n) &= \lim_{n \rightarrow \infty} \frac{|a_n|}{a_n} = \lim_{n \rightarrow \infty} \frac{|-1/n|}{-1/n} = \lim_{n \rightarrow \infty} (-1) = -1 \\ \lim_{n \rightarrow \infty} g(b_n) &= \lim_{n \rightarrow \infty} \frac{|b_n|}{b_n} = \lim_{n \rightarrow \infty} \frac{|1/n|}{1/n} = \lim_{n \rightarrow \infty} (1) = 1 \end{aligned}$$

□

Theorem 1.1.1. (Differentiable $\implies$ Continuous)

Let $I \subseteq \mathbb{R}$ be an interval, $c \in I$, and $f : I \rightarrow \mathbb{R}$ be differentiable at c . Then f is continuous at c .

Proof. It is enough to show that $\lim_{x \rightarrow c} f(x) = f(c)$ (an interval doesn't have an isolated point). Note that

$$\begin{aligned} \lim_{x \rightarrow c} (f(x) - f(c)) &= \lim_{x \rightarrow c} \left[\frac{f(x) - f(c)}{x - c} (x - c) \right] \\ &= \left[\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \right] \left[\lim_{x \rightarrow c} (x - c) \right] && \text{(ALT for Functions)} \\ &= f'(c) \cdot 0 = 0. \end{aligned}$$

So,

$$\begin{aligned}
 \lim_{x \rightarrow c} f(x) &= \lim_{x \rightarrow c} [f(x) - f(c) + f(c)] \\
 &= \lim_{x \rightarrow c} [f(x) - f(c)] + \lim_{x \rightarrow c} f(c) \\
 &= 0 + f(c) \\
 &= f(c).
 \end{aligned}$$

□

Corollary 1.1.1. If $f : I \rightarrow \mathbb{R}$ is not continuous at $c \in I$, then f is not differentiable at c .

Example 1.1.4. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \begin{cases} x^2 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$.

- (i) Prove f is continuous at 0.
- (ii) Prove f is discontinuous at all $x \neq 0$.
- (iii) Prove that f is nondifferentiable at all $x \neq 0$.
- (iv) Prove that $f'(0) = 0$.

Proof. (i) We need to show that

$$\forall \epsilon > 0 \exists \delta > 0 \text{ such that if } |x - 0| < \delta \text{ then } |f(x) - f(0)| < \epsilon$$

Let $\epsilon > 0$ be given. Our goal is to find $\delta > 0$ such that

$$\text{if } |x| < \delta \text{ then } |f(x)| < \epsilon \quad (*)$$

Informal Discussion

Note that

Case 1: if $x \notin \mathbb{Q}$ then $|f(x)| = |0| < \epsilon$ ✓

Case 2: if $x \in \mathbb{Q}$ then $|f(x)| = |x^2| = |x|^2$

So, we want to find δ such that if $|x| < \delta$, then $|x|^2 < \epsilon$. Clearly, $\delta = \sqrt{\epsilon}$ works.

We claim that $(*)$ holds with $\delta = \sqrt{\epsilon}$. See the discussion.

- (ii) Let $c \neq 0$. Our goal is to show f is discontinuous at c . By the sequential criterion for continuity, it is enough to find a sequence (a_n) such that $a_n \rightarrow c$ but $f(a_n) \not\rightarrow f(c)$. We proceed by two cases:

Case 1: $c \notin \mathbb{Q}$

$\mathbb{Q}$ is dense in $\mathbb{R}$, so there exists a sequence of rational numbers (r_n) such that $r_n \rightarrow c$. We have

$$\begin{aligned}
 \left. \begin{aligned} f(r_n) &= r_n^2 \quad \forall n \\ f(c) &= 0 \end{aligned} \right\} &\implies f(r_n) \not\rightarrow f(c) \\
 \left. \begin{aligned} r_n &\rightarrow c \\ f(r_n) &\not\rightarrow f(c) \end{aligned} \right\} &\implies f \text{ is discontinuous at } c.
 \end{aligned}$$

- (iii) Let $c \neq 0$. By (ii), f is not continuous at c . Therefore, f is not differentiable at c .

- (iv) We need to show $\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = 0$. Note that

$$\frac{f(x) - f(0)}{x - 0} = \frac{f(x)}{x}.$$

Our goal is to show:

$$\forall \epsilon > 0 \exists \delta > 0 \text{ such that if } 0 < |x - 0| < \delta \text{ then } \left| \frac{f(x)}{x} - 0 \right| < \epsilon.$$

Let $\epsilon > 0$ be given. Our goal is to find $\delta > 0$ such that

$$\text{if } 0 < |x| < \delta, \text{ then } \left| \frac{f(x)}{x} - 0 \right| < \epsilon \quad (*)$$

We claim that $(*)$ holds with $\delta = \epsilon$ (or any positive number less than ϵ). Indeed, if $x \in \mathbb{R}$ such that $0 < |x| < \delta = \epsilon$, then

Case 1: $x \notin \mathbb{Q}$

$$\left| \frac{f(x)}{x} \right| = \left| \frac{0}{x} \right| = 0 < \epsilon.$$

Case 2: $x \in \mathbb{Q}$

$$\left| \frac{f(x)}{x} \right| = \left| \frac{x^2}{x} \right| = |x| < \delta = \epsilon.$$

□

Theorem 1.1.2. (Algebraic Differentiability Theorem)

Assume $f : I \rightarrow \mathbb{R}$ and $g : I \rightarrow \mathbb{R}$ are differentiable at $c \in I$. Then

(i) $\forall k \in \mathbb{R}$, kf is differentiable at c and

$$(kf)'(c) = k \cdot f'(c)$$

(ii) $f + g$ is differentiable at c and

$$(f + g)'(c) = f'(c) + g'(c)$$

(iii) fg is differentiable at c and

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c)$$

(iv) $\frac{f}{g}$ is differentiable at c (provided $g(c) \neq 0$) and

$$\left(\frac{f}{g} \right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{[g(c)]^2}$$

Proof. Here, we will prove (ii) and (iii).

(ii)

$$\begin{aligned} \lim_{x \rightarrow c} \frac{(f + g)(x) - (f + g)(c)}{x - c} &= \lim_{x \rightarrow c} \frac{f(x) + g(x) - f(c) - g(c)}{x - c} \\ &= \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} + \lim_{x \rightarrow c} \frac{g(x) - g(c)}{x - c} \\ &= f'(c) + g'(c). \end{aligned}$$

So, $f + g$ is differentiable at c , and $(f + g)'(c) = f'(c) + g'(c)$.

(iii)

$$\begin{aligned} \lim_{x \rightarrow c} \frac{(fg)(x) - (fg)(c)}{x - c} &= \lim_{x \rightarrow c} \frac{f(x)g(x) - f(c)g(c)}{x - c} \\ &= \lim_{x \rightarrow c} \frac{f(x)g(x) - f(c)g(x) + f(c)g(x) - f(c)g(c)}{x - c} \\ &= \lim_{x \rightarrow c} \frac{(f(x) - f(c))g(x) + f(c)(g(x) - g(c))}{x - c} \\ &= \left[\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \right] \left[\lim_{x \rightarrow c} g(x) \right] + \left[\lim_{x \rightarrow c} f(c) \right] \left[\lim_{x \rightarrow c} \frac{g(x) - g(c)}{x - c} \right] \\ &= f'(c) \cdot g(c) + f(c) \cdot g'(c) \end{aligned}$$

Thus fg is differentiable at c , and $(fg)'(c) = f'(c)g(c) + f(c)g'(c)$.

□

Theorem 1.1.3. (Chain Rule)

Let $I_1 \subseteq \mathbb{R}$ and $I_2 \subseteq \mathbb{R}$ be two intervals. Suppose $f : I_1 \rightarrow \mathbb{R}$ and $g : I_2 \rightarrow \mathbb{R}$ such that $f(I_1)$ is contained in I_2 , f is differentiable at $c \in I_1$, and g is differentiable at $f(c) \in I_2$. Then the function $g \circ f : I_1 \rightarrow \mathbb{R}$ is differentiable at $c \in I_1$, and

$$(g \circ f)'(c) = g'(f(c)) \cdot f'(c).$$

Informal Discussion

The following is an incorrect proof of the theorem:

$$\begin{aligned} \lim_{x \rightarrow c} \frac{(g \circ f)(x) - (g \circ f)(c)}{x - c} &= \lim_{x \rightarrow c} \frac{g(f(x)) - g(f(c))}{x - c} \\ &= \lim_{x \rightarrow c} \frac{g(f(x)) - g(f(c))}{f(x) - f(c)} \cdot \frac{f(x) - f(c)}{x - c} \\ &= \left[\lim_{x \rightarrow c} \frac{g(f(x)) - g(f(c))}{f(x) - f(c)} \right] \cdot \left[\lim_{x \rightarrow c} \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \right] \\ &= g'(f(c)) \cdot f'(c) \end{aligned}$$

This proof fails because even though $x \rightarrow c \implies x \neq c$, it's not necessarily the case that $f(x) \rightarrow f(c) \implies f(x) \neq f(c)$. I.e., the algebraic limit theorem for functions fails as $f(x) - f(c)$ might be zero. Dividing by $f(x) - f(c)$ is not legitimate. To see why this fails, consider the case when f is a constant function.

We instead use the following idea: Replace $\frac{g(f(x)) - g(f(c))}{f(x) - f(c)}$ with a new function $d(f(x))$ such that

- (i) $d(f(x)) = \frac{g(f(x)) - g(f(c))}{f(x) - f(c)}$ when $f(x) \neq f(c)$
- (ii) $d(f(x))$ is defined even when $f(x) = f(c)$
- (iii) $d(f(x)) \cdot \frac{f(x) - f(c)}{x - c} = \frac{g(f(x)) - g(f(c))}{x - c}$ for all $x \in I_1, x \neq c$

Proof. Let $d : I_2 \rightarrow \mathbb{R}$ be defined by

$$d(y) = \begin{cases} \frac{g(y) - g(f(c))}{y - f(c)} & y \neq f(c) \\ \lim_{y \rightarrow f(c)} \frac{g(y) - g(f(c))}{y - f(c)} = g'(f(c)) & y = f(c) \end{cases}$$

Clearly, d satisfies requirements (i) and (ii) from above.

Observation 1: d is continuous at $f(c)$. Indeed,

$$\lim_{y \rightarrow f(c)} d(y) = \lim_{y \rightarrow f(c)} \frac{g(y) - g(f(c))}{y - f(c)} = d(f(c))$$

Observation 2: For all $x \in I_1$ with $x \neq c$, we have

$$d(f(x)) \cdot \frac{f(x) - f(c)}{x - c} = \frac{g(f(x)) - g(f(c))}{x - c}$$

This is true because

Case 1: $f(x) \neq f(c)$

$$\begin{aligned} d(f(x)) \cdot \frac{f(x) - f(c)}{x - c} &= \frac{g(f(x)) - g(f(c))}{f(x) - f(c)} \cdot \frac{f(x) - f(c)}{x - c} \\ &= \frac{g(f(x)) - g(f(c))}{x - c} \end{aligned}$$

Case 2: $f(x) = f(c)$

$$\begin{aligned} LHS &= d(f(x)) \cdot \frac{f(x) - f(c)}{x - c} = d(f(c)) \cdot \frac{f(x) - f(c)}{x - c} = g'(f(c)) \cdot 0 = 0 \\ RHS &= \frac{g(f(x)) - g(f(c))}{x - c} = \frac{g(f(c)) - g(f(c))}{x - c} = 0 \end{aligned}$$

So, $LHS = RHS = 0$.

We have,

$$\begin{aligned}
 \lim_{x \rightarrow c} \frac{(g \circ f)(x) - (g \circ f)(c)}{x - c} &= \lim_{x \rightarrow c} \frac{g(f(x)) - g(f(c))}{x - c} \\
 &= \lim_{x \rightarrow c} \left[d(f(x)) \cdot \frac{f(x) - f(c)}{x - c} \right] \\
 &= \left[\lim_{x \rightarrow c} (d \circ f)(x) \right] \cdot \left[\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \right] \\
 &\stackrel{(*)}{=} (d \circ f)(c) \cdot f'(c) \\
 &= d(f(c)) \cdot f'(c) \\
 &= g'(f(c)) \cdot f'(c)
 \end{aligned}$$

So, $g \circ f$ is differentiable at c and $(g \circ f)'(c) = g'(f(c)) \cdot f'(c)$.

(*) Note that f is continuous at c and d is continuous at $f(c)$, so by composition of continuous functions we conclude that $d \circ f$ is continuous at c and

$$\lim_{x \rightarrow c} (d \circ f)(x) = (d \circ f)(c).$$

□

Example 1.1.5. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \begin{cases} x^2 \sin \frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

- (i) Prove that f is differentiable at all $x \neq 0$.
- (ii) Prove that $f'(0) = 0$
- (iii) Prove that f' is not continuous at 0.

Proof. (i) We have

$$\left. \begin{array}{l} h_1(x) = 1 \text{ is differentiable on } \mathbb{R} \\ h_2(x) = x \text{ is differentiable on } \mathbb{R} \end{array} \right\} \Rightarrow \frac{h_1(x)}{h_2(x)} = \frac{1}{x} \text{ is differentiable at all } x \neq 0$$

$$\left. \begin{array}{l} h_3(x) = \sin x \text{ is differentiable on } \mathbb{R} \\ h_4(x) = \frac{1}{x} \text{ is differentiable at all } x \neq 0 \end{array} \right\} \Rightarrow (h_3 \circ h_4)(x) = \sin \frac{1}{x} \text{ is differentiable at all } x \neq 0$$

$$\left. \begin{array}{l} h_5(x) = x^2 \text{ is differentiable on } \mathbb{R} \\ h_4(x) = \sin \frac{1}{x} \text{ is differentiable at all } x \neq 0 \end{array} \right\} \Rightarrow h_5(x) \cdot h_4(x) = x^2 \sin \frac{1}{x} \text{ is differentiable at all } x \neq 0$$

Indeed, it follows from the algebraic differentiation theorem and the chain rule that

$$\begin{aligned}
 (x^2 \sin \frac{1}{x})' &= (x^2)' \cdot \sin \frac{1}{x} + x^2 \cdot (\sin \frac{1}{x})' \\
 &= 2x \cdot \sin \frac{1}{x} + x^2 \left[(\cos \frac{1}{x}) \left(-\frac{1}{x^2} \right) \right] \\
 &= 2x \sin \frac{1}{x} - \cos \frac{1}{x}
 \end{aligned}$$

- (ii) Note that $f(0) = 0$ does not imply $f'(0) = 0$. When we want to compute f' at any point, in particular at 0, we need to pay attention to the behavior of f in a neighborhood of the point and not just the value of the function at the point. The reason is that $f'(c)$ is defined by taking $\lim_{x \rightarrow c}$.

Our goal is to show

$$\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = 0$$

Note that

$$\frac{f(x) - f(0)}{x - 0} = \frac{x^2 \sin \frac{1}{x} - 0}{x - 0} = x \sin \frac{1}{x}$$

We want to show

$$\lim_{x \rightarrow 0} \left(x \sin \frac{1}{x} \right) = 0$$

We have,

$$\left. \begin{array}{l} 0 \leq \left| x \sin \frac{1}{x} \right| \leq |x| \\ \lim_{x \rightarrow 0} 0 = 0 \\ \lim_{x \rightarrow 0} |x| = |0| = 0 \end{array} \right\} \xRightarrow{\text{SQZ Thm}} \lim_{x \rightarrow 0} \left| x \sin \frac{1}{x} \right| = 0$$

Thus $\lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$.

(iii) According to parts (i) and (ii):

$$f' : \mathbb{R} \rightarrow \mathbb{R}, \quad f'(x) = \begin{cases} 2x \sin \frac{1}{x} - \cos \frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

By the sequential criterion for continuity, it is enough to find a sequence (a_n) such that

$$a_n \rightarrow 0 \text{ but } f'(a_n) \not\rightarrow f'(0)$$

Let $a_n = \frac{1}{2n\pi}$. Clearly, $a_n \rightarrow 0$. However,

$$\begin{aligned} \lim_{n \rightarrow \infty} f'(a_n) &= \lim_{n \rightarrow \infty} \left[\frac{2}{2n\pi} \sin(2n\pi) - \cos(2n\pi) \right] \\ &= 0 - 1 \\ &\neq 0. \end{aligned}$$

□

1.2 Local Extrema

Definition 1.2.1. (Local Maximum, Local Minimum)

Let $\emptyset \neq A \subseteq (X, d)$, and let $f : A \rightarrow \mathbb{R}$.

(i) We say that f has a local maximum at $c \in A$ if

$$\exists \delta > 0 \text{ such that } \forall x \in N_\delta(c) \cap A \quad f(x) \leq f(c)$$

(ii) We say that f has a local minimum at $c \in A$ if

$$\exists \delta > 0 \text{ such that } \forall x \in N_\delta(c) \cap A \quad f(x) \geq f(c)$$

Lemma 1.2.1. (Order Limit Theorem for Functions)

Suppose $\lim_{x \rightarrow c} g(x)$ and $\lim_{x \rightarrow c} h(x)$ both exist.

(i) If $\exists \delta > 0$ such that $\forall x \in (c - \delta, c) \quad h(x) \leq g(x)$, then $\lim_{x \rightarrow c} h(x) \leq \lim_{x \rightarrow c} g(x)$

(ii) If $\exists \delta > 0$ such that $\forall x \in (c, c + \delta) \quad h(x) \leq g(x)$, then $\lim_{x \rightarrow c} h(x) \leq \lim_{x \rightarrow c} g(x)$

Proof. Here we will prove (i). The proof of (ii) is analogous. Let (a_n) be a sequence in $(c - \delta, c)$ such that $a_n \rightarrow c$. By the sequential criterion for limits of functions we have

$$a_n \rightarrow c \implies \begin{cases} \lim_{n \rightarrow \infty} g(a_n) = \lim_{x \rightarrow c} g(x) \\ \lim_{n \rightarrow \infty} h(a_n) = \lim_{x \rightarrow c} h(x) \end{cases} \quad (I)$$

Also note that

$$\begin{aligned} \forall n \quad a_n \in (c - \delta, c) &\implies \forall n \quad h(a_n) \leq g(a_n) \\ &\stackrel{\text{OLTS}}{\implies} \lim_{n \rightarrow \infty} h(a_n) \leq \lim_{n \rightarrow \infty} g(a_n) \end{aligned} \quad (II)$$

It follows from (I), (II) that $\lim_{x \rightarrow c} h(x) \leq \lim_{x \rightarrow c} g(x)$. □

Theorem 1.2.1. (Interior Extremum Theorem)

Let $I \subseteq \mathbb{R}$ be an interval, $f : I \rightarrow \mathbb{R}$ be a function and $c \in \bar{I}$. Suppose f is differentiable at c . Then

(i) If f has a local maximum at c , then $f'(c) = 0$

(ii) If f has a local minimum at c , then $f'(c) = 0$

Proof. Here, we will prove (i). The proof for (ii) is analogous. Suppose f has a local maximum at c .

(i) f has a local maximum at $c \implies \exists \delta_1$ such that $\forall x \in (c - \delta_1, c + \delta_1) \cap I \quad f(x) \leq f(c)$

(ii) c is an interior point of $I \implies \exists \delta_2$ such that $(c - \delta_2, c + \delta_2) \subseteq I$

So, if we let $\delta = \min\{\delta_1, \delta_2\}$, then

$$\forall x \in (c - \delta, c + \delta) \quad f(x) \leq f(c)$$

We have

(I) For all $x \in (c - \delta, c)$

$$\begin{aligned} \left. \begin{array}{l} x - c < 0 \\ f(x) \leq f(c) \end{array} \right\} &\implies \frac{f(x) - f(c)}{x - c} \geq 0 \\ &\stackrel{OLTF}{\implies} \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \geq \lim_{x \rightarrow c} 0 \\ &\implies f'(c) \geq 0. \end{aligned}$$

(II) For all $x \in (c, c + \delta)$

$$\left. \begin{array}{l} x - c > 0 \\ f(x) \leq f(c) \end{array} \right\} \implies \frac{f(x) - f(c)}{x - c} \leq 0$$

$$\stackrel{OLTF}{\implies} \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \leq \lim_{x \rightarrow c} 0$$

$$\implies f'(c) \leq 0.$$

It follows from (I), (II) that $f'(c) = 0$. □

Remark 1.2.1. The following are three techniques that can be used in proving the existence of a solution:

- (i) Suppose $h : [a, b] \rightarrow \mathbb{R}$ is continuous. Let α be a given real number. One way to show there exists a number c such that $h(c) = \alpha$ is as follows:

$$\text{Prove that } m \leq \alpha \leq M \text{ where } \begin{cases} m = \min\{h(x) : x \in [a, b]\} \\ M = \max\{h(x) : x \in [a, b]\} \end{cases}$$

- (ii) Suppose $g : [a, b] \rightarrow \mathbb{R}$ is differentiable. One way to prove that there exists a number c such that $g'(c) = 0$ is as follows:

Prove there is a point in (a, b) at which g has a local maximum or a local minimum

- (iii) Suppose $h : [a, b] \rightarrow \mathbb{R}$ is differentiable. Let α be a given real number. One way to prove that there exists a number c such that $h'(c) = \alpha$ is as follows:

Define $g(x) = h(x) - \alpha x$ and prove that there is a point c at which $g'(c) = 0$

Theorem 1.2.2. (Darboux's Theorem)

Suppose $f : [a, b] \rightarrow \mathbb{R}$ is differentiable such that $f'(a) < f'(b)$ (or $f'(b) < f'(a)$), and let $\alpha \in \mathbb{R}$ be such that $f'(a) < \alpha < f'(b)$ (or $f'(b) < \alpha < f'(a)$). Then

$$\exists c \in (a, b) \text{ such that } f'(c) = \alpha$$

Proof. Let $g : [a, b] \rightarrow \mathbb{R}$ be defined by $g(x) = f(x) - \alpha x$. It follows from the algebraic differentiability theorem that g is differentiable on $[a, b]$, and so it is continuous on $[a, b]$. It is enough to show that

$$\exists c \in (a, b) \text{ such that } g'(c) = 0$$

To this end, it is enough to show that $\exists c \in (a, b)$ at which g has a local minimum. We have

$$\left. \begin{array}{l} g \text{ is continuous on } [a, b] \\ [a, b] \text{ is compact} \end{array} \right\} \implies g \text{ attains its minimum on } [a, b]$$

Let $\hat{c}$ be a point at which g attains a minimum. In what follows we will show that $\hat{c} \in (a, b)$ and so it can be used as the c that we were looking for. Note that (since $g'(x) = f'(x) - \alpha$)

$$\begin{aligned} g'(a) &= f'(a) - \alpha < 0 \\ g'(b) &= f'(b) - \alpha > 0 \end{aligned}$$

Claim 1: $\hat{c} \neq a$

Assume for contradiction that $\hat{c} = a$. Then

$$\forall x \in [a, b] \quad g(x) \geq g(a)$$

so,

$$\forall x \in [a, b] \quad \begin{cases} g(x) - g(a) \geq 0 \\ x - a > 0 \end{cases}$$

Thus

$$\forall x \in (a, b) \quad \frac{g(x) - g(a)}{x - a} \geq 0$$

Thus

$$\lim_{x \rightarrow c} \frac{g(x) - g(a)}{x - a} \geq \lim_{x \rightarrow a} 0$$

That is, $g'(a) \geq 0$. This contradicts the fact that $g'(a) < 0$.

Claim 2: $\hat{c} \neq b$

Assume for contradiction that $\hat{c} = b$. In a similar manner to claim 1:

$$\begin{aligned} \forall x \in [a, b] \quad g(x) \geq g(b) &\implies \forall x \in [a, b] \quad \begin{cases} g(x) - g(b) \geq 0 \\ x - b < 0 \end{cases} \\ &\implies \forall x \in [a, b] \quad \frac{g(x) - g(b)}{x - b} \leq 0 \end{aligned}$$

Thus,

$$\lim_{x \rightarrow c} \frac{g(x) - g(b)}{x - b} \leq \lim_{x \rightarrow b} 0$$

That is,

$$g'(b) \leq 0.$$

This contradicts the fact that $g'(b) > 0$.

Example 1.2.1. Does there exist a differentiable function $f : [-1, 1] \rightarrow \mathbb{R}$ whose derivative is $H : [-1, 1] \rightarrow \mathbb{R}$ defined by

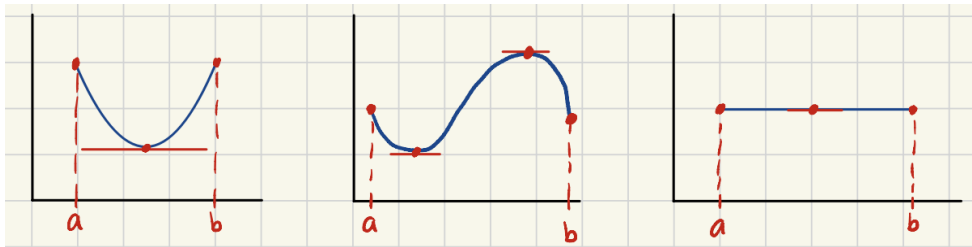
$$H(x) = \begin{cases} 1 & 0 \leq x \leq 1 \\ 0 & -1 \leq x \leq 0 \end{cases} ?$$

No! H does not have the intermediate value property. So, it cannot be the derivative of any differentiable function.

The following are some geometric conjectures involving the derivative of a function.

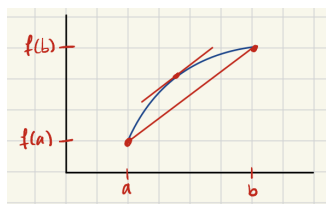
Conjecture 1.2.1.

Suppose $f : [a, b] \rightarrow \mathbb{R}$ is differentiable. Suppose $f(a) = f(b)$. Then there exists a point $c \in (a, b)$ at which the tangent line is horizontal. I.e., there exists $c \in (a, b)$ such that $f'(c) = 0$.



Conjecture 1.2.2.

Suppose $f : [a, b] \rightarrow \mathbb{R}$ is differentiable. Then there exists a point $c \in (a, b)$ at which the tangent line is parallel to the line through the endpoints $(a, f(a))$ and $(b, f(b))$. I.e., there exists $c \in (a, b)$ such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.



Conjecture 1.2.3.

Suppose $\vec{r} : [a, b] \rightarrow \mathbb{R}^2$, $\vec{r}(t) = (f(t), g(t))$ is a differentiable path in $\mathbb{R}^2$. Then there exists a point $\vec{r}(c)$ on the curve at which the tangent line is parallel to the line through the endpoints $\vec{r}(a)$ and $\vec{r}(b)$. Let's

try to find a mathematical formula for this statement:

- *) The direction vector for the tangent line at the point $\vec{r}(c)$: $\vec{r}'(c) = (f'(c), g'(c))$
- *) The direction vector for the line through the endpoints: $(f(b) - f(a), g(b) - g(a))$

So, assuming these vectors are nonzero, the claim of the conjecture can be described mathematically as

$$\exists c \exists \lambda \in \mathbb{R} \setminus \{0\} \text{ such that } (f'(c), g'(c)) = \lambda (f(b) - f(a), g(b) - g(a))$$

Note that

$$\begin{aligned} (f'(c), g'(c)) &= \lambda (f(b) - f(a), g(b) - g(a)) \\ \implies \begin{cases} f'(c) = \lambda (f(b) - f(a)) \\ g'(c) = \lambda (g(b) - g(a)) \end{cases} \\ \implies \lambda f'(c) [g(b) - g(a)] &= \lambda g'(c) [f(b) - f(a)] \\ \implies f'(c) [f(b) - f(a)] &= g'(c) [g(b) - g(a)] \end{aligned}$$

1.3 Mean Value Theorems

We now study three theorems that make the previous geometric observations precise.

Theorem 1.3.1. (Rolle's Theorem)

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. Let f be differentiable on (a, b) . Suppose $f(a) = f(b)$. Then there exists a point $c \in (a, b)$ such that $f'(c) = 0$.

Proof. It is enough to show that there exists a point $c \in (a, b)$ at which f has a local maximum or a local minimum. We have

$$\left. \begin{array}{l} f \text{ is continuous} \\ [a, b] \text{ is compact} \end{array} \right\} \xRightarrow{EVT} f \text{ attains its maximum and minimum on } [a, b]$$

We consider two cases:

Case 1: Both $\max_{a \leq x \leq b} f(x)$ and $\min_{a \leq x \leq b} f(x)$ occur at the endpoints.

In this case, it follows from the assumption $f(a) = f(b)$ that $\max_{a \leq x \leq b} f(x) = \min_{a \leq x \leq b} f(x)$. So, f is a constant function on $[a, b]$. Hence

$$\forall x \in [a, b] \quad f'(x) = 0$$

So, we may choose c to be any point we like in (a, b) .

Case 2: Either $\max_{a \leq x \leq b} f(x)$ or $\min_{a \leq x \leq b} f(x)$ occurs at a point $c \in (a, b)$.

It follows from the interior extreme value theorem that $f'(c) = 0$.

□

Theorem 1.3.2. (Mean Value Theorem)

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous and let f be differentiable on (a, b) . Then there exists $c \in (a, b)$ such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Proof. Let $g : [a, b] \rightarrow \mathbb{R}$ be defined by

$$g(x) = f(x) - \frac{f(b) - f(a)}{b - a}x$$

Note that

*) Algebraic continuity theorem $\implies g$ is continuous on $[a, b]$

*) Algebraic differentiability theorem $\implies g$ is differentiable on (a, b)

$$*) \quad g(a) = f(a) - \frac{f(b) - f(a)}{b - a}a = \frac{bf(a) - af(a) - af(b) + af(a)}{b - a} = \frac{bf(a) - af(a)}{b - a}$$

$$*) \quad g(b) = f(b) - \frac{f(b) - f(a)}{b - a}b = \frac{bf(a) - af(b)}{b - a}$$

g is continuous on $[a, b]$, differentiable on (a, b) , and $g(a) = g(b)$. By Rolle's theorem,

$$\exists c \in (a, b) \text{ such that } g'(c) = 0$$

Note that $g'(x) = f'(x) - \frac{f(b) - f(a)}{b - a}$, so

$$\begin{aligned} g'(c) = 0 &\iff f'(c) - \frac{f(b) - f(a)}{b - a} = 0 \\ &\iff f'(c) = \frac{f(b) - f(a)}{b - a} \end{aligned}$$

□

Theorem 1.3.3. (Generalized Mean Value Theorem)

Let $f : [a, b] \rightarrow \mathbb{R}$ and $g : [a, b] \rightarrow \mathbb{R}$ be continuous functions that are differentiable on (a, b) . Then there

exists a point $c \in (a, b)$ such that

$$f'(c) [g(b) - g(a)] = g'(c) [f(b) - f(a)]$$

Proof. Let $h(x) = [f(b) - f(a)]g(x) - [g(b) - g(a)]f(x)$. It follows from the assumptions, the algebraic continuity theorem, and the algebraic differentiability theorem that h is continuous on $[a, b]$ and differentiable on (a, b) . Therefore, by the mean value theorem,

$$\exists c \in (a, b) \text{ such that } h'(c) = \frac{h(b) - h(a)}{b - a} \quad (*)$$

Note that

$$\begin{aligned} h(b) &= [f(b) - f(a)]g(b) - [g(b) - g(a)]f(b) \\ &= f(b)g(b) - f(a)g(b) - g(b)f(b) + g(a)f(b) \\ &= g(a)f(b) - f(a)g(b) \\ h(a) &= [f(b) - f(a)]g(a) - [g(b) - g(a)]f(a) \\ &= f(b)g(a) - f(a)g(a) - g(b)f(a) + g(a)f(a) \\ &= f(b)g(a) - g(b)f(a) \end{aligned}$$

So $h(a) = h(b)$. Hence it follows from $(*)$ that $\exists c \in (a, b)$ such that $h'(c) = 0$. Now note that

$$\begin{aligned} h'(x) &= [f(b) - f(a)]g'(x) - [g(b) - g(a)]f'(x) \\ \implies h'(c) &= [f(b) - f(a)]g'(c) - [g(b) - g(a)]f'(c) \end{aligned}$$

Therefore,

$$\exists c \in (a, b) \text{ such that } [f(b) - f(a)]g'(c) - [g(b) - g(a)]f'(c) = 0$$

That is,

$$\exists c \in (a, b) \text{ such that } [f(b) - f(a)]g'(c) = [g(b) - g(a)]f'(c)$$

□

Remark 1.3.1. If g' is never zero in (a, b) , then we may rewrite the claim of general mean value theorem as follows:

$$\exists c \in (a, b) \text{ such that } \frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

Theorem 1.3.4.

Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$ be differentiable such that $f'(x) = 0 \forall x \in I$. Then f is a constant function on I , that is, there exists $k \in \mathbb{R}$ such that $\forall x \in I, f(x) = k$.

Proof. Let $x, y \in I$ with $x < y$. It is enough to show that $f(x) = f(y)$. To this end, we will apply the mean value theorem to f on the interval $[x, y]$:

$$\begin{aligned} \exists c \in (x, y) \text{ such that } f'(c) &= \frac{f(y) - f(x)}{y - x} \\ \implies 0 &= \frac{f(y) - f(x)}{y - x} \\ \implies 0 &= f(y) - f(x) \\ \implies f(x) &= f(y) \end{aligned}$$

□

Remark 1.3.2. Consider $f : A \rightarrow \mathbb{R}$ where $A = (-1, 0) \cup (2, 3)$ and $f(x) = \begin{cases} 1 & x \in (-1, 0) \\ -1 & x \in (2, 3) \end{cases}$. Then $\forall x \in A, f'(x) = 0$, but f is not a constant function on A . The theorem above doesn't apply since A is not an interval.

Theorem 1.3.5.

Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \rightarrow \mathbb{R}$ and $g : I \rightarrow \mathbb{R}$ be differentiable such that $f'(x) = g'(x) \forall x \in I$. Then there exists $k \in \mathbb{R}$ such that $\forall x \in I, f(x) = g(x) + k$.

Proof. Let $h = f - g$. We have

$$\begin{aligned} \forall x \in I \quad h'(x) &= (f - g)'(x) = f'(x) - g'(x) = 0 \\ &\stackrel{1.3.4}{\implies} \exists k \in \mathbb{R} \text{ such that } \forall x \in I \quad h(x) = k \\ &\implies \exists k \in \mathbb{R} \text{ such that } \forall x \in I \quad f(x) - g(x) = k \\ &\implies \exists k \in \mathbb{R} \text{ such that } \forall x \in I \quad f(x) = g(x) + k \end{aligned}$$

□

Theorem 1.3.6.

Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$ be differentiable. Then

$$(i) \quad f \text{ is increasing} \iff \forall c \in I \quad f'(c) \geq 0$$

$$(ii) \quad f \text{ is decreasing} \iff \forall c \in I \quad f'(c) \leq 0$$

Proof.

Here, we will prove (i). The proof of (ii) is analogous.

($\implies$) Suppose f is increasing on I . Let $c \in I$. Note that for all $x \in I, x \neq c$ we have $\frac{f(x) - f(c)}{x - c} \geq 0$. Indeed,

$$\begin{aligned} \text{if } x > c \text{ then } \begin{cases} x - c > 0 \\ f(x) \geq f(c) \end{cases} &\implies \frac{f(x) - f(c)}{x - c} \geq 0 \\ \text{if } x < c \text{ then } \begin{cases} x - c < 0 \\ f(x) \leq f(c) \end{cases} &\implies \frac{f(x) - f(c)}{x - c} \geq 0 \end{aligned}$$

It follows from the order limit theorem for functions that

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \geq \lim_{x \rightarrow c} 0$$

Hence, $f'(c) \geq 0$ as desired.

($\impliedby$) Suppose $\forall c \in I \quad f'(c) \geq 0$. Let $x_1, x_2 \in I$ with $x_1 < x_2$. It is enough to show that $f(x_1) \leq f(x_2)$. To this end, we apply the mean value theorem to the function f on $[x_1, x_2]$:

$$\exists c \in (x_1, x_2) \text{ such that } f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

So, $f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$. Thus $f(x_2) - f(x_1) \geq 0$, that is, $f(x_1) \leq f(x_2)$ as desired. □

Theorem 1.3.7. (L'Hôpital's Rule)

Let $I \subseteq \mathbb{R}$ be an interval, and $a \in I$. Let $f : [a, b] \rightarrow \mathbb{R}$ and $g : [a, b] \rightarrow \mathbb{R}$ be continuous. Suppose f and g are differentiable at all points in $I \setminus \{a\}$ and $f(a) = g(a) = 0$, $g'(x) \neq 0 \forall x \in I \setminus \{a\}$ and $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = L \in \mathbb{R}$.

Then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = L$.

Proof. Our goal is to show that

$$\forall \epsilon > 0 \exists \delta > 0 \text{ such that if } 0 < |x - a| < \delta \text{ (with } x \in I) \text{ then } \left| \frac{f(x)}{g(x)} - L \right| < \epsilon$$

Let $\epsilon > 0$. Our goal is to find $\delta > 0$ such that

$$\text{if } 0 < |x - a| < \delta \text{ (with } x \in I) \text{ then } \left| \frac{f(x)}{g(x)} - L \right| < \epsilon \quad (*)$$

Since by assumption $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = L$, for the given $\epsilon > 0$, there exists $\hat{\delta} > 0$ such that

$$\text{if } 0 < |x - a| < \hat{\delta} \text{ (with } x \in I \text{) then } \left| \frac{f'(x)}{g'(x)} - L \right| < \epsilon$$

We claim that this $\hat{\delta}$ satisfies (*). The reason is as follows:

Suppose $x \in I$ such that $0 < |x - a| < \hat{\delta}$. In what follows we will show that $\left| \frac{f(x)}{g(x)} - L \right| < \epsilon$. We consider two cases:

Case 1: $x > a$ ($x \in (a, a + \hat{\delta})$)

We apply the general mean value theorem to f and g on the interval $[a, x]$:

$$\exists c \in (a, x) \text{ such that } \frac{f'(c)}{g'(c)} = \frac{f(x) - f(a)}{g(x) - g(a)}$$

Since $f(a) = g(a) = 0$, we conclude that

$$\frac{f'(c)}{g'(c)} = \frac{f(x)}{g(x)}$$

It follows that

$$\left| \frac{f(x)}{g(x)} - L \right| = \left| \frac{f'(c)}{g'(c)} - L \right| < \epsilon$$

(the latter inequality is true because $0 < |c - a| \leq |x - a| \leq \hat{\delta}$)

Case 2: $x < a$ ($x \in (a - \hat{\delta}, a)$)

We apply the general mean value theorem to f and g on $[x, a]$:

$$\exists c \in (x, a) \text{ such that } \frac{f'(c)}{g'(c)} = \frac{f(a) - f(x)}{g(a) - g(x)}$$

Since $f(a) = g(a) = 0$, we conclude that

$$\frac{f'(c)}{g'(c)} = \frac{f(x)}{g(x)}$$

It follows that

$$\left| \frac{f(x)}{g(x)} - L \right| = \left| \frac{f'(c)}{g'(c)} - L \right| < \epsilon$$

(the latter inequality is true because $0 < |c - a| \leq |x - a| \leq \hat{\delta}$)

□

1.4 Taylor Polynomials

Consider $f : I \rightarrow \mathbb{R}$ given by $f(x) = (x - x_0)^k$. What do the n derivatives look like? If $n = 0$, then by the mean value theorem we have

$$\left. \begin{array}{l} f : I \rightarrow \mathbb{R} \text{ is differentiable} \\ f(x_0) = 0 \end{array} \right\} \implies f(x) = f'(c)(x - x_0)$$

Observation 1. Let k be a natural number. Let x_0 be a fixed number.

$$*) \frac{d}{dx} [(x - x_0)^k] = k(x - x_0)^{k-1}$$

$$*) \frac{d^2}{dx^2} [(x - x_0)^k] = \frac{d}{dx} [k(x - x_0)^{k-1}] = k(k-1)(x - x_0)^{k-2}$$

$$*) \frac{d^3}{dx^3} [(x - x_0)^k] = \frac{d}{dx} [k(k-1)(x - x_0)^{k-2}] = k(k-1)(k-2)(x - x_0)^{k-3}$$

$\vdots$

$$*) \frac{d^k}{dx^k} [(x - x_0)^k] = k(k-1) \dots (2)(1)(x - x_0)^{k-k} = k!$$

If $j < k$

$$*) \frac{d^j}{dx^j} [(x - x_0)^k] = k(k-1) \dots (k-(j-1))(x - x_0)^{k-j}$$

Thus we have

$$\frac{d^j}{dx^j} [(x - x_0)^k] = \begin{cases} k(k-1) \dots (k-j+1)(x - x_0)^{k-j} & \text{if } j < k \\ k! & \text{if } j = k \\ 0 & \text{if } j > k \end{cases}$$

$$\frac{d^j}{dx^j} [(x - x_0)^k] \Big|_{x=x_0} = \begin{cases} 0 & \text{if } j < k \\ k! & \text{if } j = k \\ 0 & \text{if } j > k \end{cases}$$

Theorem 1.4.1. (Corollary of the General Mean Value Theorem)

Let $I \subseteq \mathbb{R}$ be an open interval, $x_0 \in I$, and $n \in \mathbb{N} \cup \{0\}$. Let $f : I \rightarrow \mathbb{R}$ have $n+1$ derivatives. Suppose $f^{(k)}(x_0) = 0 \quad \forall 0 \leq k \leq n$. Then for each point $x \neq x_0$ in the interval I , there exists a point c_{x,x_0} strictly between x and x_0 such that

$$f(x) = \frac{f^{(n+1)}(c_{x,x_0})}{(n+1)!} (x - x_0)^{n+1}$$

Proof. Here we will prove the claim for the case where $x > x_0$. The proof for $x < x_0$ is completely analogous. Let $g : I \rightarrow \mathbb{R}$ be defined by $g(t) = (t - x_0)^{n+1}$. Note that

$$g^{(k)}(x_0) = 0 \quad \forall 0 \leq k \leq n$$

$$g^{(n+1)}(t) = (n+1)! \quad \forall t \in I$$

Now, we apply the general mean value theorem to f and g on the interval $[x_0, x]$:

$$\begin{aligned} \exists x_1 \in (x_0, x) \text{ such that } \frac{f'(x_1)}{g'(x_1)} &= \frac{f(x) - f(x_0)}{g(x) - g(x_0)} \\ \implies \frac{f'(x_1)}{g'(x_1)} &= \frac{f(x)}{g(x)} \end{aligned} \tag{I}$$

Next, we apply the general mean value theorem to f' and g' on the interval $[x_0, x_1]$:

$$\begin{aligned} \exists x_2 \in (x_0, x_1) \text{ such that } \frac{f''(x_2)}{g''(x_2)} &= \frac{f'(x_1) - f'(x_0)}{g'(x_1) - g'(x_0)} \\ \implies \frac{f''(x_2)}{g''(x_2)} &= \frac{f'(x_1)}{g'(x_1)} \\ \stackrel{(I)}{\implies} \frac{f''(x_2)}{g''(x_2)} &= \frac{f(x)}{g(x)} \end{aligned}$$

Continuing in this way, we will obtain $x_{n+1} \in (x_0, x)$ such that

$$\frac{f^{(n+1)}(x_{n+1})}{g^{(n+1)}(x_{n+1})} = \frac{f(x)}{g(x)}$$

So,

$$\frac{f^{(n+1)}(x_{n+1})}{(n+1)!} = \frac{f(x)}{(x-x_0)^{n+1}}$$

Thus

$$f(x) = \frac{f^{(n+1)}(x_{n+1})}{(n+1)!} (x-x_0)^{n+1}$$

(We can use x_{n+1} as the c we were looking for) □

Question: What are the nicest functions that we know? Which functions are the easiest to work with?

Answer: Polynomials

General Question: Given a function f , is it possible to find a "good" approximation for f among polynomials?

Setup:

- *) Let I be a nonempty open interval in $\mathbb{R}$
- *) Let n be a nonnegative integer
- *) Suppose $f : I \rightarrow \mathbb{R}$ has n derivatives and $x_0 \in I$
- *) Suppose that we want to use the values

$$f(x_0), f'(x_0), \dots, f^{(n)}(x_0)$$

to construct a polynomial approximation for f

What is the best we could hope for? Find a polynomial such that

$$\begin{aligned} p(x_0) &= f(x_0) \\ p'(x_0) &= f'(x_0) \\ &\vdots \\ p^{(n)}(x_0) &= f^{(n)}(x_0) \end{aligned}$$

Observation 2. Let x_0 be a fixed real number. A general polynomial of degree at most n can be expressed in powers of $(x - x_0)$ in the form

$$p(x) = c_0 + c_1(x - x_0) + \dots + c_n(x - x_0)^n$$

Example 1.4.1. Consider $p(x) = x^2 - 3x - 1$. Let $x_0 = 1$. We can express $p(x)$ in powers of $x - 1$:

$$\begin{aligned} p(x) &= x^2 - 3x - 1 = [(x-1) + 1]^2 - 3[(x-1) + 1] - 1 \\ &= (x-1)^2 + 2(x-1) + 1 - 3(x-1) - 3 - 1 \\ &= (x-1)^2 - (x-1) - 3 \end{aligned}$$

Theorem 1.4.2. (Uniqueness of the Approximating Polynomial)

Let $I \subseteq \mathbb{R}$ be an open interval and $n \in \mathbb{N}$. Suppose $f : I \rightarrow \mathbb{R}$ has n derivatives and $x_0 \in I$. Then there exists a unique polynomial $p(x)$ of degree at most n such that

$$\forall 0 \leq l \leq n \quad p^{(l)}(x_0) = f^{(l)}(x_0), \text{ with } \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k$$

Proof. Let $p(x)$ be a general polynomial of degree at most n :

$$p(x) = c_0 + c_1(x - x_0) + \dots + c_n(x - x_0)^n$$

Our goal is to show that

$$\text{If } \forall 0 \leq l \leq n \quad p^{(l)}(x_0) = f^{(l)}(x_0) \text{ then } p(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$$

Note that $p(x_0) = c_0$. Also, for $1 \leq l \leq n$ we have

$$\begin{aligned} p^{(l)}(x) &= \frac{d^l}{dx^l} \left[c_0 + \sum_{k=1}^n c_k (x - x_0)^k \right] \\ &= \frac{d^l}{dx^l} \left[\sum_{k=1}^n c_k (x - x_0)^k \right] \\ &= \sum_{k=1}^n c_k \frac{d^l}{dx^l} [(x - x_0)^k] \end{aligned}$$

Hence,

$$p^{(l)}(x_0) = \sum_{k=1}^n c_k \frac{d^l}{dx^l} [(x - x_0)^k] \Big|_{x=x_0} = c_l \cdot l!$$

Therefore,

$$\forall 1 \leq l \leq n \quad p^{(l)}(x_0) = c_l \cdot l!$$

We conclude that

$$\begin{aligned} p \text{ agrees with } f \text{ to order } n \text{ at } x_0 &\iff \begin{cases} p(x_0) = f(x_0) \\ p^{(l)}(x_0) = f^{(l)}(x_0) \quad \forall 1 \leq l \leq n \end{cases} \\ &\iff \begin{cases} c_0 = f(x_0) \\ l!c_l = f^{(l)}(x_0) \quad \forall 1 \leq l \leq n \end{cases} \\ &\iff \begin{cases} c_0 = f(x_0) \\ c_l = \frac{f^{(l)}(x_0)}{l!} \quad \forall 1 \leq l \leq n \end{cases} \\ &\iff p(x) = \sum_{k=0}^n c_k (x - x_0)^k \\ &= c_0 + \sum_{k=1}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k \\ &= f(x_0) + \sum_{k=1}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k \\ &= \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k \end{aligned}$$

□

Note . n^{th} Taylor Polynomial centered at 0 is called n^{th} Maclaurin Polynomial

Big Lesson: There is exactly one polynomial of degree at most n that satisfies

$$\begin{aligned} p(x_0) &= f(x_0) \\ p'(x_0) &= f'(x_0) \\ &\vdots \\ p^{(n)}(x_0) &= f^{(n)}(x_0) \end{aligned}$$

This polynomial is called the n^{th} Taylor polynomial for f centered at x_0 , and is given by

$$T_{n,x_0}(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \dots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$$

Theorem 1.4.3. (Taylor's Theorem with Lagrange Remainder)

Let $I \subseteq \mathbb{R}$ be an open interval, $x_0 \in I$, and $n \in \mathbb{N} \cup \{0\}$. Let $f : I \rightarrow \mathbb{R}$ have $n + 1$ derivatives. Then for each point $x \neq x_0$ in I , there is a point c strictly between x and x_0 such that

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + \frac{f^{(n+1)}(c)}{(n+1)!} (x - x_0)^{n+1}$$

Remark 1.4.1. Note that clearly the above equality holds at $x = x_0$ too (for any value of c). Recall that for any fixed number R , $\lim_{n \rightarrow \infty} \frac{R^{n+1}}{(n+1)!} = 0$, however $f^{(n+1)}(c)$ may become very large.

Proof. Let $F_{n,x_0} = f(x) - T_{n,x_0}(x)$. Our goal is to show that

$$R_{n,x_0} = \frac{f^{(n+1)}(c)}{(n+1)!} (x - x_0)^{n+1}$$

for some c between x and x_0 . Note that

$$\left. \begin{array}{l} f \text{ has } n+1 \text{ derivatives} \\ (i) \ T_{n,x_0} \text{ is a polynomial of degree } n, \text{ so it has } n+1 \text{ derivatives} \\ R_{n,x_0} = f - T \end{array} \right\} \implies R_{n,x_0} \text{ has } n+1 \text{ derivatives}$$

$$(ii) \ \forall 0 \leq k \leq n \ R_{n,x_0}^{(k)}(x_0) = f^{(k)}(x_0) - T_{n,x_0}^{(k)}(x_0) = 0$$

(i), (ii), Theorem 1.4.1 $\implies$ For each point $x \neq x_0$ in I , we have

$$R_{n,x_0}(x) = \frac{R_{n,x_0}^{(n+1)}(c)}{(n+1)!} (x - x_0)^{n+1} \text{ for some } c \text{ strictly between } x \text{ and } x_0 \quad (I)$$

Now note that

$$R_{n,x_0}^{(n+1)}(c) = f^{(n+1)}(c) - T_{n,x_0}^{(n+1)}(c) = f^{(n+1)}(c) \quad (II)$$

$$(I), (II) \implies R_{n,x_0} = \frac{f^{(n+1)}(c)}{(n+1)!} (x - x_0)^{(n+1)}$$

□

Chapter 2

Integration

2.1 The Riemann-Stieltjes Integral

Definition 2.1.1. (Almost Disjoint Intervals)

We say two intervals I and J are almost disjoint if either $I \cap J = \emptyset$ or $I \cap J$ has a single element.

Definition 2.1.2. (Partition)

First Viewpoint: A partition P of an interval $[a, b]$ is a finite set of points in $[a, b]$ that include both a and b . We always list the points of a partition $P = \{x_0, \dots, x_n\}$ in increasing order:

$$x_0 = a < x_1 < x_2 < \dots < x_n = b$$

Second Viewpoint: A partition P of an interval $[a, b]$ is a finite collection of almost disjoint (nonempty) compact intervals whose union is $[a, b]$:

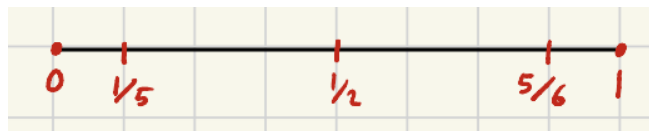
$$P = I_1, \dots, I_n \text{ where } I_1 = [x_0, x_1], I_2 = [x_1, x_2], \dots, I_n = [x_{n-1}, x_n] \text{ (with } x_0 = a, x_n = b)$$

Example 2.1.1.

$$P = \left\{ 0, \frac{1}{5}, \frac{1}{2}, \frac{5}{6}, 1 \right\}$$

$$P = \left\{ [0, \frac{1}{5}], [\frac{1}{5}, \frac{1}{2}], [\frac{1}{2}, \frac{5}{6}], [\frac{5}{6}, 1] \right\}$$

are both partitions of $[0, 1]$.



Notation . Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded function. Let $P = \{x_0 = a, \dots, x_n = b\}$ be a partition of $[a, b]$. We let

$$m = \inf \{f(x) : x \in [a, b]\}$$

$$M = \sup \{f(x) : x \in [a, b]\}$$

$$\forall 1 \leq k \leq n \quad m_k = \inf \{f(x) : x \in [x_{k-1}, x_k]\}$$

$$\forall 1 \leq k \leq n \quad M_k = \sup \{f(x) : x \in [x_{k-1}, x_k]\}$$

The existence of m, M, m_k, M_k as real numbers is guaranteed due to the assumption that f is bounded on $[a, b]$.

Remark 2.1.1. Suppose A and B are nonempty, bounded sets in $\mathbb{R}$ such that $A \subseteq B$. Then

- (i) $\inf A \leq \sup A$
- (ii) $\inf B \leq \sup B$
- (iii) $\sup A \leq \sup B$
- (iv) $\inf A \geq \inf B$

As a result

$$\forall 1 \leq k \leq n \quad m \leq m_k \leq M_k \leq M$$

Remark 2.1.2. If $f : [a, b] \rightarrow \mathbb{R}$ is continuous, then f attains its maximum and minimum over each compact subinterval. In this case,

$$\begin{aligned} m_k &= \min \{f(x) : x \in [x_{k-1}, x_k]\} \\ M_k &= \max \{f(x) : x \in [x_{k-1}, x_k]\} \end{aligned}$$

Definition 2.1.3. (Lower Sum, Upper Sum)

Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded and let $\alpha : [a, b] \rightarrow \mathbb{R}$ be increasing. Let $P = \{x_0, \dots, x_n\}$ be a partition of $[a, b]$. Let $\Delta\alpha_k = \alpha(x_k) - \alpha(x_{k-1})$.

- (i) The lower Riemann-Stieltjes sum of f (R.S. sum of f) with respect to α for the partition P is defined by

$$L(f, \alpha, P) = \sum_{k=1}^n m_k (\alpha(x_k) - \alpha(x_{k-1})) = \sum_{k=1}^n m_k \Delta\alpha_k$$

- (ii) The upper Riemann-Stieltjes sum of f (R.S sum of f) with respect α for the partition P is defined by

$$U(f, \alpha, P) = \sum_{k=1}^n M_k (\alpha(x_k) - \alpha(x_{k-1})) = \sum_{k=1}^n M_k \Delta\alpha_k$$

Note . Note that

$$m(\alpha(b) - \alpha(a)) \leq L(f, \alpha, P) \leq U(f, \alpha, P) \leq M(\alpha(b) - \alpha(a))$$

Indeed,

$$\begin{aligned} L(f, \alpha, P) &= \sum_{k=1}^n m_k \Delta\alpha_k \geq \sum_{k=1}^n m(\alpha(x_k) - \alpha(x_{k-1})) \\ &= m \sum_{k=1}^n (\alpha(x_k) - \alpha(x_{k-1})) \\ &= m [\alpha(x_1) - \alpha(x_0) + \alpha(x_2) - \alpha(x_1) + \dots + \alpha(x_n) - \alpha(x_{n-1})] \\ &= m [\alpha(x_n) - \alpha(x_0)] \\ &= m [\alpha(b) - \alpha(a)] \end{aligned}$$

Similarly,

$$\begin{aligned} U(f, \alpha, P) &= \sum_{k=1}^n M_k \Delta\alpha_k \leq \sum_{k=1}^n M(\alpha(x_k) - \alpha(x_{k-1})) \\ &= M \sum_{k=1}^n (\alpha(x_k) - \alpha(x_{k-1})) \\ &= M [\alpha(x_1) - \alpha(x_0) + \alpha(x_2) - \alpha(x_1) + \dots + \alpha(x_n) - \alpha(x_{n-1})] \\ &= M [\alpha(x_n) - \alpha(x_0)] \\ &= M [\alpha(b) - \alpha(a)] \end{aligned}$$

Notation . $\Pi[a, b]$, or Π for short, denotes the collection of all the possible partitions of $[a, b]$.

Definition 2.1.4. (Upper Riemann-Stieltjes Integral, Lower Riemann-Stieltjes Integral)

Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded and $\alpha : [a, b] \rightarrow \mathbb{R}$ be increasing.

- (i) The upper Riemann-Stieltjes integral of f with respect to α (on $[a, b]$) is defined by

$$U(f, \alpha) = \overline{\int_a^b f d\alpha} = \inf\{U(f, \alpha, P) : P \in \Pi\}$$

(Note that the set $\{U(f, \alpha, P) : P \in \Pi\}$ of all upper sums is bounded below by $m(\alpha(b) - \alpha(a))$, so the infimum is a real number)

- (ii) The lower Riemann-Stieltjes integral of f with respect to α (on $[a, b]$) is defined by

$$L(f, \alpha) = \underline{\int_a^b f d\alpha} = \sup\{L(f, \alpha, P) : P \in \Pi\}$$

Definition 2.1.5. (Riemann-Stieltjes Integrable Function)

Let $\alpha : [a, b] \rightarrow \mathbb{R}$ be an increasing function. A function $f : [a, b] \rightarrow \mathbb{R}$ is said to be Riemann-Stieltjes integrable (on $[a, b]$) with respect to α if

- (i) f is bounded
(ii) $L(f, \alpha) = U(f, \alpha)$

In this case, the Riemann-Stieltjes integral of f with respect to α , denoted by

$$\int_a^b f d\alpha \text{ or } \int_a^b f(x) d\alpha \text{ or } \int_{[a, b]} f d\alpha$$

is the common value of $L(f, \alpha)$ and $U(f, \alpha)$. That is,

$$\int_a^b f d\alpha = L(f, \alpha) = U(f, \alpha)$$

Example 2.1.2. Let c be a fixed real number. Prove that the constant function $f(x) = c$ on $[a, b]$ is R.S. integrable and

$$\int_a^b f d\alpha = c(b - a)$$

Proof. For any partition $P = \{x_0, \dots, x_n\}$ of $[a, b]$ we have

$$\begin{aligned} \forall 1 \leq k \leq n, m_k &= \inf\{f(x) : x \in [x_{k-1}, x_k]\} = \inf\{c\} = c \\ M_k &= \sup\{f(x) : x \in [x_{k-1}, x_k]\} = \sup\{c\} = c \end{aligned}$$

Therefore,

$$\begin{aligned} L(f, \alpha, P) &= \sum_{k=1}^n m_k (\alpha(x_k) - \alpha(x_{k-1})) \\ &= \sum_{k=1}^n c (\alpha(x_k) - \alpha(x_{k-1})) \\ &= c \sum_{k=1}^n (\alpha(x_k) - \alpha(x_{k-1})) \\ &= c[\alpha(b) - \alpha(a)] \end{aligned}$$

and

$$\begin{aligned}
 U(f, \alpha, P) &= \sum_{k=1}^n M_k(\alpha(x_k) - \alpha(x_{k-1})) \\
 &= \sum_{k=1}^n c(\alpha(x_k) - \alpha(x_{k-1})) \\
 &= c \sum_{k=1}^n (\alpha(x_k) - \alpha(x_{k-1})) \\
 &= c[\alpha(b) - \alpha(a)]
 \end{aligned}$$

Hence,

$$\begin{aligned}
 L(f, \alpha) &= \sup\{L(f, \alpha, P) : P \in \Pi\} = \sup\{c(\alpha(b) - \alpha(a))\} = c(\alpha(b) - \alpha(a)) \\
 U(f, \alpha) &= \inf\{U(f, \alpha, P) : P \in \Pi\} = \inf\{c(\alpha(b) - \alpha(a))\} = c(\alpha(b) - \alpha(a))
 \end{aligned}$$

Since $L(f, \alpha) = U(f, \alpha) = c(\alpha(b) - \alpha(a))$, we conclude that f is R.S. integrable with respect to α and

$$\int_a^b f d\alpha = c[\alpha(b) - \alpha(a)]$$

□

Example 2.1.3. Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded and let $\alpha : [a, b] \rightarrow \mathbb{R}$ be a constant function ($\alpha(x) = c$). Prove that $\int_a^b f d\alpha = 0$.

Proof. For any partition $P = \{x_0, \dots, x_n\}$ of $[a, b]$,

$$\begin{aligned}
 L(f, \alpha, P) &= \sum_{k=1}^n m_k [\alpha(x_k) - \alpha(x_{k-1})] = \sum_{k=1}^n m_k \cdot 0 = 0 \\
 U(f, \alpha, P) &= \sum_{k=1}^n M_k [\alpha(x_k) - \alpha(x_{k-1})] = \sum_{k=1}^n M_k \cdot 0 = 0
 \end{aligned}$$

Therefore,

$$\begin{aligned}
 L(f, \alpha) &= \sup\{L(f, \alpha, P) : P \in \Pi\} = 0 \\
 U(f, \alpha) &= \inf\{U(f, \alpha, P) : P \in \Pi\} = 0
 \end{aligned}$$

So $L(f, \alpha) = U(f, \alpha) = 0 \implies \int_a^b f d\alpha = 0$.

□

Example 2.1.4. Let $f : [a, b] \rightarrow \mathbb{R}$ be defined by

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$$

Prove that

- (i) if $\alpha : [a, b] \rightarrow \mathbb{R}$ is constant, then $\int_a^b f d\alpha = 0$ (proved in the last example)
- (ii) if $\alpha : [a, b] \rightarrow \mathbb{R}$ is increasing and $\alpha(a) \neq \alpha(b)$, then $f \notin R_\alpha[a, b]$

Proof. (ii) For any partition $P = \{x_0, \dots, x_n\}$ of $[a, b]$ we have

$$\begin{aligned}
 \forall 1 \leq k \leq n \quad m_k &= \inf\{f(x) : x \in [x_{k-1}, x_k]\} = \inf\{0, 1\} = 0 \\
 M_k &= \sup\{f(x) : x \in [x_{k-1}, x_k]\} = \sup\{0, 1\} = 1
 \end{aligned}$$

Therefore

$$L(f, \alpha, P) = \sum_{k=1}^n m_k [\alpha(x_k) - \alpha(x_{k-1})] = 0 \cdot [\alpha(b) - \alpha(a)] = 0$$

$$U(f, \alpha, P) = \sum_{k=1}^n M_k [\alpha(x_k) - \alpha(x_{k-1})] = 1 \cdot [\alpha(b) - \alpha(a)] = \alpha(b) - \alpha(a)$$

Hence

$$\left. \begin{aligned} L(f, \alpha) &= \sup\{L(f, \alpha, P) : P \in \Pi\} = 0 \\ U(f, \alpha) &= \inf\{U(f, \alpha, P) : P \in \Pi\} = \alpha(b) - \alpha(a) \end{aligned} \right\} \implies L(f, \alpha) \neq U(f, \alpha) \implies f \notin R_\alpha[a, b]$$

□

Definition 2.1.6. (Refinement of a Partition)

First Viewpoint: A partition $Q = \{z_0, \dots, z_m\}$ of $[a, b]$ is a refinement of a partition $P = \{x_0, \dots, x_n\}$ of $[a, b]$ if $P \subseteq Q$. That is, if Q contains all points of P .

Second Viewpoint: A partition $Q = J_1, \dots, J_m$ of $[a, b]$ is a refinement of a partition $P = \{I_0, \dots, I_n\}$ of $[a, b]$ if every interval I_k of P is an almost disjoint union of one or more intervals of Q .

Example 2.1.5. Consider the following partitions of $[0, 1]$:

$$P = \{0, \frac{1}{2}, 1\}$$

$$Q = \{0, \frac{1}{3}, \frac{1}{2}, 1\}$$

Then Q is a refinement of P since $P \subseteq Q$.

Remark 2.1.3. Let P and Q be any two partitions of $[a, b]$. Then $P \cup Q$ will be a refinement of both P and Q because $P \subseteq P \cup Q$ and $Q \subseteq P \cup Q$. So, any two partitions of $[a, b]$ have a common refinement.

Example 2.1.6. Consider the following partitions of $[0, 1]$:

$$P = \{0, \frac{1}{2}, 1\}$$

$$Q = \{0, \frac{1}{3}, \frac{2}{3}, 1\}$$

P is not a refinement of Q and Q is not a refinement of P , but

$$P \cup Q = \{0, \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, 1\}$$

is a refinement of both P and Q .

Definition 2.1.7. (Size of a Partition)

Let $P = \{x_0, \dots, x_n\}$ be a partition of $[a, b]$. The size of P , denoted $||P||$, is defined by

$$||P|| = \max \{|x_k - x_{k-1}| : 1 \leq k \leq n\}$$

$$= \max \{|x_1 - x_0|, |x_2 - x_1|, \dots, |x_n - x_{n-1}|\}$$

2.2 9 Useful Theorems of Integrability

Theorem 2.2.1. (Inequalities of Refinements)

Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded and let $\alpha : [a, b] \rightarrow \mathbb{R}$ be increasing. Let P be a partition of $[a, b]$ and let Q be a refinement of P . Then

$$(i) \quad L(f, \alpha, P) \leq L(f, \alpha, Q)$$

$$(ii) \quad U(f, \alpha, P) \geq U(f, \alpha, Q)$$

Proof. Here, we will prove (i). The proof of (ii) is completely analogous. We proceed by induction on $l = \text{card}(Q \setminus P)$ (the number of points in $Q \setminus P$). Let $P = \{x_0, \dots, x_n\}$.

Base Case: If $l = 0$, then

$$\left. \begin{array}{l} P \subseteq Q \\ \text{card } Q = \text{card } P \end{array} \right\} \implies P = Q \implies L(f, \alpha, P) = L(f, \alpha, Q)$$

If $l = 1$, then Q has exactly one extra point. Let's call this point z , so $\{z\} = Q \setminus P$. Note that

$$\left. \begin{array}{l} z \in [a, b] \\ P \text{ is a partition of } [a, b] \end{array} \right\} \implies \exists 1 \leq i \leq n \text{ such that } z \in (x_{i-1}, x_i)$$

Let

$$\begin{aligned} m'_i &= \inf \{f(x) : x \in [x_{i-1}, z]\} \\ m''_i &= \inf \{f(x) : x \in [z, x_i]\} \end{aligned}$$

Recall that if $A \subseteq B$, then $\inf A \geq \inf B$. Hence $m'_i \geq m_i$ and $m''_i \geq m_i$ (where $m_i = \inf \{f(x) : x \in [x_{i-1}, x_i]\}$). We have

$$\begin{aligned} L(f, \alpha, P) &= \sum_{k=1}^n m_k [\alpha(x_k) - \alpha(x_{k-1})] \\ &= \left[\sum_{k=1, k \neq i}^n m_k (\alpha(x_k) - \alpha(x_{k-1})) \right] + m_i (\alpha(x_i) - \alpha(x_{i-1})) \\ &= \left[\sum_{k=1, k \neq i}^n m_k (\alpha(x_k) - \alpha(x_{k-1})) \right] + m_i (\alpha(z) - \alpha(x_{i-1})) + m_i (\alpha(x_i) - \alpha(z)) \\ &\leq \left[\sum_{k=1, k \neq i}^n m_k (\alpha(x_k) - \alpha(x_{k-1})) \right] + m'_i (\alpha(z) - \alpha(x_{i-1})) + m''_i (\alpha(x_i) - \alpha(z)) \\ &= L(f, \alpha, Q) \end{aligned}$$

so

$$L(f, \alpha, P) \leq L(f, \alpha, Q)$$

Inductive Step: Suppose the claim is true for $l = r \geq 1$. We want to show the claim holds for $l = r + 1$.

Suppose $\text{card}(Q \setminus P) = r + 1$. Let

$$Q \setminus P = \{z_1, \dots, z_r, z_{r+1}\}$$

Let

$$\hat{Q} = P \cup \{z_1, \dots, z_r\}$$

We have

$$L(f, \alpha, P) \stackrel{\text{hypoth.}}{\leq} L(f, \alpha, \hat{Q}) \stackrel{\text{base case}}{\leq} L(f, \alpha, Q)$$

So,

$$L(f, \alpha, P) \leq L(f, \alpha, Q)$$

□

Theorem 2.2.2. (Lower Sums are Smaller than Upper Sums)

Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded and $\alpha : [a, b] \rightarrow \mathbb{R}$ be increasing. Let P_1 and P_2 be any two partitions of $[a, b]$. Then $L(f, \alpha, P_1) \leq U(f, \alpha, P_2)$.

Proof. Let $Q = P_1 \cup P_2$ be the common refinement of P_1 and P_2 . We have

$$L(f, \alpha, P_1) \leq L(f, \alpha, Q) \leq U(f, \alpha, Q) \leq U(f, \alpha, P_2)$$

□

Theorem 2.2.3. (The Lower R.S. Integral is less than the Upper R.S. Integral)

Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded and $\alpha : [a, b] \rightarrow \mathbb{R}$ be increasing. Then $L(f, \alpha) \leq U(f, \alpha)$.

Proof. Note that if A and B are nonempty subsets of $\mathbb{R}$ such that $\forall a \in A \quad \forall b \in B \quad a \leq b$, then $\sup A \leq \inf B$. Let $A = \{L(f, \alpha, P) : P \in \Pi\}$ and $B = \{U(f, \alpha, P) : P \in \Pi\}$. By Thm 2.2.2, $a \leq b$ for every $a \in A$ and $b \in B$. So, it follows that $\sup A \leq \sup B$, that is $L(f, \alpha) \leq U(f, \alpha)$. □

Theorem 2.2.4. (Cauchy Criterion for R.S. Integrability)

Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded and let $\alpha : [a, b] \rightarrow \mathbb{R}$ be increasing.

$$f \in \mathbb{R}_\alpha[a, b] \iff \forall \epsilon > 0 \quad \exists P_\epsilon \in \Pi \text{ such that } U(f, \alpha, P_\epsilon) - L(f, \alpha, P_\epsilon) < \epsilon.$$

Proof. ($\Leftarrow$) Suppose $f : [a, b]$ is bounded and $\alpha : [a, b] \rightarrow \mathbb{R}$ is increasing, and

$$\forall \epsilon > 0 \quad \exists P_\epsilon \in \Pi \text{ such that } U(f, \alpha, P_\epsilon) - L(f, \alpha, P_\epsilon) < \epsilon$$

We want to show $f \in R_\alpha[a, b]$. That is, we want to show $L(f, \alpha) = U(f, \alpha)$. Since $U(f, \alpha) - L(f, \alpha) \geq 0$, it is enough to show that

$$\forall \epsilon > 0 \quad U(f, \alpha) - L(f, \alpha) < \epsilon$$

Let $\epsilon > 0$ be given. By assumption, there exists $P_\epsilon \in \Pi$ such that

$$U(f, \alpha, P_\epsilon) - L(f, \alpha, P_\epsilon) < \epsilon$$

We have

$$\begin{aligned} U(f, \alpha) &= \inf \{U(f, \alpha, P) : P \in \Pi\} \leq U(f, \alpha, P_\epsilon) \\ L(f, \alpha) &= \sup \{L(f, \alpha, P) : P \in \Pi\} \geq L(f, \alpha, P_\epsilon) \end{aligned}$$

Hence,

$$L(f, \alpha, P_\epsilon) \leq L(f, \alpha) \stackrel{\text{Thm 2.2.3}}{\leq} U(f, \alpha) \leq U(f, \alpha, P_\epsilon)$$

The interval $[L(f, \alpha), U(f, \alpha)]$ is contained in the interval $[L(f, \alpha, P_\epsilon), U(f, \alpha, P_\epsilon)]$. Thus

$$U(f, \alpha) - L(f, \alpha) \leq U(f, \alpha, P_\epsilon) - L(f, \alpha, P_\epsilon) < \epsilon$$

as desired.

($\Leftarrow$) Suppose $f \in R_\alpha[a, b]$. Our goal is to show

$$\forall \epsilon > 0 \quad \exists P_\epsilon \in \Pi \text{ such that } U(f, \alpha, P_\epsilon) - L(f, \alpha, P_\epsilon) < \epsilon$$

Let $\epsilon > 0$ be given. Our goal is to find $P_\epsilon \in \Pi$ such that

$$U(f, \alpha, P_\epsilon) - L(f, \alpha, P_\epsilon) < \epsilon$$

Note that

$$\begin{aligned} U(f, \alpha) &= \inf \{U(f, \alpha, P) : P \in \Pi\} \implies \exists P_1 \in \Pi \text{ such that } U(f, \alpha, P_1) < U(f, \alpha) + \frac{\epsilon}{2} \\ L(f, \alpha) &= \sup \{L(f, \alpha, P) : P \in \Pi\} \implies \exists P_2 \in \Pi \text{ such that } L(f, \alpha) - \frac{\epsilon}{2} < L(f, \alpha, P_2) \end{aligned}$$

Let $P_\epsilon = P_1 \cup P_2$. We claim that this partition can be used to prove the claim.

$$L(f, \alpha) - \frac{\epsilon}{2} < L(f, \alpha, P_2) \leq L(f, \alpha, P_\epsilon) \leq U(f, \alpha, P_\epsilon) \leq U(f, \alpha, P_1) < U(f, \alpha) + \frac{\epsilon}{2}$$

Thus

$$\begin{aligned}
 U(f, \alpha, P_\epsilon) - L(f, \alpha, P_\epsilon) &< [(U(f, \alpha) + \epsilon/2) - (L(f, \alpha) - \epsilon/2)] \\
 &= U(f, \alpha) - L(f, \alpha) + \epsilon \\
 &= 0 + \epsilon \\
 &= \epsilon
 \end{aligned}$$

□

Theorem 2.2.5. (Refinement and Tag Stability for R.S. Integrals)

Suppose $f : [a, b] \rightarrow \mathbb{R}$ is bounded and $\alpha : [a, b] \rightarrow \mathbb{R}$ is increasing. Let $\epsilon > 0$ and let $P = \{x_0, \dots, x_n\}$ be a partition of $[a, b]$ such that $U(f, \alpha, P) - L(f, \alpha, P) < \epsilon$. Then

(i) If Q is a refinement of P , then $U(f, \alpha, Q) - L(f, \alpha, Q) < \epsilon$.

(ii) If $\forall 1 \leq k \leq n$ t_k and s_k are arbitrary points in $[x_{k-1}, x_k]$, then

$$\sum_{k=1}^n |f(s_k) - f(t_k)| \Delta\alpha_k < \epsilon$$

(iii) If $f \in R_\alpha[a, b]$ and $\forall 1 \leq k \leq n$ $s_k \in [x_{k-1}, x_k]$, then

$$\left| \sum_{k=1}^n f(s_k) \Delta\alpha_k - \int_a^b f d\alpha \right| < \epsilon$$

Proof. (i) We have

$$L(f, \alpha, P) \leq L(f, \alpha, Q) \leq U(f, \alpha, Q) \leq U(f, \alpha, P)$$

Therefore,

$$U(f, \alpha, Q) - L(f, \alpha, Q) \leq U(f, \alpha, P) - L(f, \alpha, P) < \epsilon$$

(ii) For each $1 \leq k \leq n$, we have

$$\begin{aligned}
 m_k &\leq f(s_k) \leq M_k \\
 m_k &\leq f(t_k) \leq M_k \iff -M_k \leq -f(t_k) \leq -m_k
 \end{aligned}$$

so

$$\begin{aligned}
 m_k - M_k &\leq f(s_k) - f(t_k) \leq M_k - m_k \\
 \implies -(M_k - m_k) &\leq f(s_k) - f(t_k) \leq M_k - m_k \\
 \implies |f(s_k) - f(t_k)| &\leq M_k - m_k
 \end{aligned}$$

Hence,

$$\sum_{k=1}^n |f(s_k) - f(t_k)| \Delta\alpha_k \leq \sum_{k=1}^n (M_k - m_k) \Delta\alpha_k = U(f, \alpha, P) - L(f, \alpha, P) < \epsilon$$

(iii) $\forall 1 \leq k \leq n$ $m_k \leq f(s_k) \leq M_k$ so,

$$\sum_{k=1}^n m_k \Delta\alpha_k \leq \sum_{k=1}^n f(s_k) \Delta\alpha_k \leq \sum_{k=1}^n M_k \Delta\alpha_k$$

Therefore,

$$L(f, \alpha, P) \leq \sum_{k=1}^n f(s_k) \Delta\alpha_k \leq U(f, \alpha, P)$$

Also, note that

$$L(f, \alpha, P) \leq \int_a^b f d\alpha \leq U(f, \alpha, P)$$

Hence,

$$\left| \sum_{k=1}^n f(s_k) \Delta\alpha_k - \int_a^b f d\alpha \right| \leq U(f, \alpha, P) - L(f, \alpha, P) < \epsilon$$

□

Remark 2.2.1. Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded, $\alpha : [a, b] \rightarrow \mathbb{R}$ be increasing, and $P = \{x_0, \dots, x_n\}$ be a partition of $[a, b]$.

- * If for every $1 \leq k \leq n$ we choose a point $t_k \in [x_{k-1}, x_k]$, then we call the pair $(P, (t_k)_{1 \leq k \leq n})$ a tagged partition.
- * We sometimes refer to $(t_k)_{1 \leq k \leq n}$ as a tag of a partition.
- * A sum of the form

$$S(f, \alpha, (P, (t_k)_{1 \leq k \leq n})) = \sum_{k=1}^n f(t_k) \Delta \alpha_k$$

is called a Riemann-Stieltjes sum of f with respect to α (The Riemann-Stieltjes sum of f with respect to α for the tagged partition $(P, (t_k)_{1 \leq k \leq n})$).

- * As a direct consequence of Theorem 2.2.5 and the Cauchy Criterion for Riemann-Stieltjes integrability, we have

$$\begin{aligned} & f \in R_\alpha[a, b] \\ \implies & \forall \epsilon > 0 \quad \exists P_\epsilon \in \Pi \text{ such that } \forall P_\epsilon \subseteq P \quad \forall \text{tag}(t_k) \text{ of } P \quad \left| S(f, \alpha, (P, (t_k)_{1 \leq k \leq n})) - \int_a^b f d\alpha \right| < \epsilon \end{aligned}$$

The converse is true in the following sense:

Suppose $f : [a, b] \rightarrow \mathbb{R}$ is bounded and $\alpha : [a, b] \rightarrow \mathbb{R}$ is increasing. Then the following two statements are equivalent:

- (i) $f \in R_\alpha[a, b]$
- (ii) There exists a number A such that

$$\forall \epsilon > 0 \quad \exists P_\epsilon \in \Pi \text{ such that } \forall P_\epsilon \subseteq P \quad \forall \text{tag}(t_k) \text{ of } P \quad |S(f, \alpha, (P, (t_k)_{1 \leq k \leq n})) - A| < \epsilon$$

Lemma 2.2.1. Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded function. Let $P = \{x_0, \dots, x_n\}$ be a partition of $[a, b]$. Then

$$\forall 1 \leq k \leq n \quad \sup \{|f(s) - f(t)| : s, t \in [x_{k-1}, x_k]\} = M_k - m_k$$

Proof. Let $k \in \{1, \dots, n\}$. We need to show

- (i) $\forall s, t \in [x_{k-1}, x_k] \quad |f(s) - f(t)| \leq M_k - m_k$
- (ii) $\forall \epsilon > 0 \quad \exists \hat{s}, \hat{t} \in [x_{k-1}, x_k] \text{ such that } M_k - m_k < |f(\hat{s}) - f(\hat{t})|$

We proved (i) in our discussion of theorem 2.2.5. It remains to show (ii):

Let $\epsilon > 0$ be given. Then

$$\begin{aligned} m_k = \inf \{f(t) : [x_{k-1}, x_k]\} & \implies \exists \hat{t} \in [x_{k-1}, x_k] \text{ such that } f(\hat{t}) < m_k + \frac{\epsilon}{2} \implies -m_k - \frac{\epsilon}{2} < -f(\hat{t}) \\ M_k = \sup \{f(t) : [x_{k-1}, x_k]\} & \implies \exists \hat{s} \in [x_{k-1}, x_k] \text{ such that } M_k - \frac{\epsilon}{2} < f(\hat{s}) \end{aligned}$$

Therefore, adding the inequalities gives us

$$M_k - m_k - \epsilon < f(\hat{s}) - f(\hat{t}) \leq |f(\hat{s}) - f(\hat{t})|$$

□

Theorem 2.2.6. (Continuity Implies Integrability)

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous and let $\alpha : [a, b] \rightarrow \mathbb{R}$ be increasing. Then $f \in R_\alpha[a, b]$.

Proof. First, note that

$$\left. \begin{array}{l} f : [a, b] \rightarrow \mathbb{R} \text{ is continuous} \\ [a, b] \text{ is compact} \end{array} \right\} \xRightarrow{\text{EVT}} f \text{ is bounded on } [a, b]$$

Now, according to the Cauchy Criterion of R.S. integrability, it is enough to show that

$$\forall \epsilon > 0 \quad \exists P \in \Pi \text{ such that } U(f, \alpha, P) - L(f, \alpha, P) < \epsilon$$

Let $\epsilon > 0$ be given. Our goal is to find $P \in \Pi$ such that

$$U(f, \alpha, P) - L(f, \alpha, P) < \epsilon \quad (*)$$

We have

$$\left. \begin{array}{l} f \text{ is continuous on } [a, b] \\ [a, b] \text{ is compact} \end{array} \right\} \implies f \text{ is uniformly continuous on } [a, b]$$

So, for the given *epsilon*,

$$\exists \delta \text{ such that } \forall s, t \in [a, b] \quad |s - t| < \delta \implies |f(s) - f(t)| < \frac{\epsilon}{2[\alpha(b) - \alpha(a) + 1]}$$

Let P be any partition of $[a, b]$ such that $\|P\| < \delta$. We claim that $(*)$ holds for such a P . Indeed, $\forall k \in \{1, \dots, n\} \quad \forall s, t \in [x_{k-1}, x_k]$ we have

$$|s - t| < \delta \implies |f(t) - f(s)| < \frac{\epsilon}{2[\alpha(b) - \alpha(a) + 1]}$$

Thus

$$\sup_{s, t \in [x_{k-1}, x_k]} |f(s) - f(t)| \leq \frac{\epsilon}{2[\alpha(b) - \alpha(a) + 1]}$$

Hence, by lemma 2.2.1,

$$M_k - m_k \leq \frac{\epsilon}{2[\alpha(b) - \alpha(a) + 1]}$$

Therefore,

$$\begin{aligned} U(f, \alpha, P) - L(f, \alpha, P) &= \sum_{k=1}^n M_k - m_k \Delta \alpha_k \\ &\leq \sum_{k=1}^n \frac{\epsilon}{2[\alpha(b) - \alpha(a) + 1]} \Delta \alpha_k \\ &= \frac{\epsilon}{2[\alpha(b) - \alpha(a) + 1]} [\alpha(b) - \alpha(a)] \\ &\leq \frac{\epsilon}{2} \\ &< \epsilon \end{aligned}$$

□

Example 2.2.1. $f : [0, 10] \rightarrow \mathbb{R}$ defined by $f(x) = \begin{cases} \frac{\sin x}{x} & x > 0 \\ 1 & x = 0 \end{cases}$ is continuous.

$\alpha : [0, 10] \rightarrow \mathbb{R}$ defined by $\alpha(x) = \lfloor x \rfloor$ is an increasing function.

Hence, by theorem 2.2.6,

$$\int_0^{10} f(x) d\lfloor x \rfloor \text{ exists}$$

Lemma 2.2.2. Let $\alpha : [a, b] \rightarrow \mathbb{R}$ be increasing and continuous and let $\alpha(a) < \alpha(b)$. Then for each $n \in \mathbb{N}$, there exists a partition $P = \{x_0, \dots, x_n\}$ such that

$$\forall 1 \leq k \leq n \quad \Delta \alpha_k = \alpha(x_k) - \alpha(x_{k-1}) = \frac{\alpha(b) - \alpha(a)}{n}$$

Proof. Let $n \in \mathbb{N}$. Divide the interval $[\alpha(a), \alpha(b)]$ into n subintervals of equal length $\frac{\alpha(b) - \alpha(a)}{n}$. For each $1 < k < n$, we have $y_k \in (\alpha(a), \alpha(b))$, hence by the intermediate value theorem

$$\exists x_k \in (a, b) \text{ such that } y_k = \alpha(x_k)$$

Since α is increasing, we have $a < x_1 < x_2 < \dots < x_{n-1} < b$. Now, if we define $x_0 = a$ and $x_n = b$, then $P = \{x_0, \dots, x_n\}$ will be a partition of $[a, b]$ such that

$$\forall 1 \leq k \leq n \quad \delta\alpha_k = \alpha(x_k) - \alpha(x_{k-1}) = y_k - y_{k-1} = \frac{\alpha(b) - \alpha(a)}{n}$$

□

Theorem 2.2.7. (Monotone Functions are Riemann-Stieltjes Integrable)

Suppose $\alpha : [a, b] \rightarrow \mathbb{R}$ is increasing and continuous.

- (i) If $f : [a, b] \rightarrow \mathbb{R}$ is increasing, then $f \in R_\alpha[a, b]$
- (ii) If $f : [a, b] \rightarrow \mathbb{R}$ is decreasing, then $f \in R_\alpha[a, b]$

Proof. We prove (i) ((ii) is analogous). First note that

$$\forall x \in [a, b] \quad f(a) \leq f(x) \leq f(b) \implies f \text{ is bounded on } [a, b]$$

If $\alpha(a) = \alpha(b)$, then we previously proved that $f \in R_\alpha[a, b]$ and $\int_a^b f d\alpha = 0$. So it remains to prove the claim for the case where $\alpha(a) \neq \alpha(b)$. According to the Cauchy Criterion for integrability, in order to show $f \in R_\alpha[a, b]$, it is enough to show that

$$\forall \epsilon > 0 \quad \exists P \in \Pi \text{ such that } U(f, \alpha, P) - L(f, \alpha, P) < \epsilon.$$

Let $\epsilon > 0$ be given. Our goal is to find $P \in \Pi$ such that

$$U(f, \alpha, P) - L(f, \alpha, P) < \epsilon$$

Choose $n \in \mathbb{N}$ large enough so that

$$\frac{\alpha(b) - \alpha(a)}{n} [f(b) - f(a)] < \epsilon.$$

Let $\tilde{P} = \{x_0, \dots, x_n\}$ be a partition of $[a, b]$ such that

$$\begin{aligned} \forall 1 \leq k \leq n \quad \delta\alpha_k &= \alpha(x_k) - \alpha(x_{k-1}) \\ &= \frac{\alpha(b) - \alpha(a)}{n} \end{aligned}$$

We claim that $\tilde{P}$ can be used as the P we are looking for. Indeed,

$$\begin{aligned} U(f, \alpha, \tilde{P}) - L(f, \alpha, \tilde{P}) &= \sum_{k=1}^n (M_k - m_k) \delta\alpha_k \\ &= \frac{\alpha(b) - \alpha(a)}{n} \sum_{k=1}^n (M_k - m_k) \\ &= \frac{\alpha(b) - \alpha(a)}{n} \sum_{k=1}^n [f(x_k) - f(x_{k-1})] \quad (\text{for all } k \text{ } M_k = \sup f(x) = f(x_k) \text{ and } m_k = f(x_{k-1})) \\ &= \frac{\alpha(b) - \alpha(a)}{n} [f(b) - f(a)] \\ &< \epsilon \end{aligned}$$

as desired. □

Example 2.2.2. $\alpha : [0, 2] \rightarrow \mathbb{R}$ given by $\alpha(x) = x^2$ is continuous and increasing and $f : [0, 2] \rightarrow \mathbb{R}$ given by $f(x) = \lfloor x^2 \rfloor$ is increasing. By theorem 2.2.7, $\int_0^2 \lfloor x^2 \rfloor d(x^2)$ exists.

Remark 2.2.2. Recall that to show $f \in R_\alpha[a, b]$, it is enough to show

$$\forall \epsilon > 0 \quad \exists P \in \Pi \text{ such that } U(f, \alpha, P) - L(f, \alpha, P) < \epsilon,$$

Also, note that

$$U(f, \alpha, P) - L(f, \alpha, P) = \sum_{k=1}^n (M_k - m_k) \delta \alpha_k.$$

So, we would like to make $\sum (M_k - m_k) \delta \alpha_k$ small.

*) Sometimes, this can be done by ensuring that $M_k - m_k$ is small

$$M_k - m_k < \epsilon \implies \sum (M_k - m_k) \delta \alpha_k < \epsilon \sum \delta \alpha_k = \epsilon [\alpha(b) - \alpha(a)]$$

*) Sometimes, this can be done by making $\delta \alpha_k$ small

$$\sum (M_k - m_k) \delta \alpha_k \stackrel{\text{Lemma 2.2.1}}{\leq} \sum 2\tilde{M} \delta \alpha_k = 2\tilde{M} \sum \delta \alpha_k$$

where $\tilde{M} = \sup |f(x)|$.

Theorem 2.2.8. (Finite Discontinuities Implies R.S. Integrability)

Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded and suppose f has finitely many discontinuities: $y_1 < y_2 < \dots < y_n$. Suppose $\alpha : [a, b] \rightarrow \mathbb{R}$ is increasing and α is continuous at $y_1 < y_2 < \dots < y_n$. Then $f \in R_\alpha[a, b]$.

Proof. According to the Cauchy criterion for Riemann-Stieltjes integrability, it is enough to show that

$$\forall \epsilon > 0 \quad \exists P \in \Pi \text{ such that } U(f, \alpha, P) - L(f, \alpha, P) < \epsilon$$

Let $\epsilon > 0$ be given. Our goal is to find P such that

$$U(f, \alpha, P) - L(f, \alpha, P) < \epsilon$$

Let $\tilde{M} = \sup_{a \leq x \leq b} |f(x)|$. Let $\hat{\epsilon} = \frac{\epsilon}{[\alpha(b) - \alpha(a)] + 2\tilde{M} + 1}$. We make two claims:

Claim 1: There exists finitely many disjoint intervals $[u_1, v_1], \dots, [u_m, v_m]$ such that

- (i) $\forall 1 \leq j \leq m \quad y_j \in [u_j, v_j]$.
- (ii) $\forall 1 \leq j \leq m$ if $y_j \notin \{a, b\}$ then $y_j \in (u_j, v_j)$.
- (iii) $\forall 1 \leq j \leq m \quad \alpha(v_j) - \alpha(u_j) < \frac{\hat{\epsilon}}{m}$ and so $\sum_{j=1}^m \alpha(v_j) - \alpha(u_j) < \epsilon$.

Claim 2: Let $K = [a, b] \setminus \bigcup_{j=1}^m (u_j, v_j)$. Then f is uniformly continuous on K .

We will prove claim 1 and claim 2 shortly, for now assume they hold. By claim 2,

$$\exists \delta > 0 \text{ such that } \forall s, t \in K \text{ if } |s - t| < \delta \text{ then } |f(s) - f(t)| < \hat{\epsilon}.$$

Now, we form a partition $\tilde{P}$ of $[a, b]$ as follows:

- (i) $\forall 1 \leq j \leq m \quad u_j, v_j \in \tilde{P}$.
- (ii) $\forall 1 \leq j \leq m$ no point of the segment (u_j, v_j) is in $\tilde{P}$.
- (iii) If $1 \leq k \leq n$ is such that $x_{k-1} \notin \{u_1, \dots, u_m\}$, then we choose x_k such that $x_k - x_{k-1} < \delta$.

We claim that $\tilde{P}$ can be used as the P that we were looking for. Indeed, let

$$A = \{k : x_{k-1} \notin \{u_1, \dots, u_m\}\}, B = \{1, \dots, n\} \setminus A.$$

Note that

*) if $k \in A$, then $x_k - x_{k-1} < \delta$, so

$$\begin{aligned} & \forall s, t \in [x_{k-1}, x_k] \quad |s - t| < \epsilon \\ \implies & \forall s, t \in [x_{k-1}, x_k] \quad |f(s) - f(t)| < \hat{\epsilon} \\ \implies & \sup_{s, t \in [x_{k-1}, x_k]} |f(s) - f(t)| \leq \hat{\epsilon} \\ \implies & M_k - m_k \leq \hat{\epsilon} \end{aligned}$$

*) if $k \in B$, then

$$M_k - m_k = \sup_{s, t \in [x_{k-1}, x_k]} |f(s) - f(t)| \leq 2\tilde{M}$$

Therefore,

$$\begin{aligned} U(f, \alpha, \tilde{P}) - L(f, \alpha, \tilde{P}) &= \sum_{k=1}^n (M_k - m_k) \Delta \alpha_k \\ &= \sum_{k \in A} (M_k - m_k) \Delta \alpha_k + \sum_{k \in B} (M_k - m_k) \Delta \alpha_k \\ &\leq \sum_{k \in A} \hat{\epsilon} \Delta \alpha_k + \sum_{k \in B} 2\tilde{M} \Delta \alpha_k \\ &\leq \tilde{\epsilon} \sum_{k=1}^n \Delta \alpha_k + 2\tilde{M} \sum_{k \in B} \Delta \alpha_k \\ &\leq \tilde{\epsilon} [\alpha(b) - \alpha(a)] + 2\tilde{M} \hat{\epsilon} \\ &= [\alpha(b) - \alpha(a) + 2\tilde{M}] \hat{\epsilon} \\ &< \epsilon \end{aligned}$$

Proof of Claim 1: Since for each $q \leq j \leq m$ α is continuous at y_j , we can choose $\delta_j > 0$ such that

$$\text{if } |y - y_j| < \delta_j \text{ then } |\alpha(y) - \alpha(y_j)| < \frac{\hat{\epsilon}}{2M}.$$

Now let

$$\tilde{\delta} = \frac{\min}{4} \{ \delta_1, \delta_2, \dots, \delta_m, y_2 - y_1, y_3 - y_2, y_4 - y_3, \dots, y_m - y_{m-1} \}.$$

For each $1 \leq j \leq m$, we define

- (i) If $y_j \notin \{a, b\}$, then $[u_j, v_j] = [y_j - \tilde{\delta}, y_j + \tilde{\delta}]$.
- (ii) If $y_j = a$, then $[u_j, v_j] = [a, a + \tilde{\delta}]$.
- (iii) If $y_j = b$, then $[u_j, v_j] = [b - \tilde{\delta}, b]$.

Clearly, these intervals satisfy all the requirements. In particular,

$$\alpha(v_j) - \alpha(u_j) = |\alpha(v_j) - \alpha(u_j)| \leq |\alpha(v_j) - \alpha(y_j)| + |\alpha(y_j) - \alpha(u_j)| < \frac{\hat{\epsilon}}{2M} + \frac{\hat{\epsilon}}{2M} = \frac{\hat{\epsilon}}{M}$$

Proof of Claim 2: Note that $\bigcup_{j=1}^m (u_j, v_j)$ is open. Hence,

$$K = [a, b] \setminus \bigcup_{j=1}^m (u_j, v_j) = [a, b] \cap \left[\bigcup_{j=1}^m (u_j, v_j) \right]^c \implies K \text{ is closed.}$$

$$\left. \begin{array}{l} K \subseteq [a, b] \\ K \text{ is closed} \\ [a, b] \text{ is compact} \end{array} \right\} \implies K \text{ is compact.}$$

Since $f : K \rightarrow \mathbb{R}$ is continuous and K is compact, we can conclude that f is uniformly continuous on K . □

Theorem 2.2.9. (Applying Continuous Functions Preserves Integrability)

Suppose $f \in R_\alpha[a, b]$ such that $\forall x \in [a, b] \quad m \leq f(x) \leq M$ and suppose $\varphi : [m, M] \rightarrow \mathbb{R}$ is continuous. Then $\varphi \circ f \in R_\alpha[a, b]$.

Proof. First, note that a composition of bounded functions is bounded, so $h = \varphi \circ f$ is bounded. According to the Cauchy criterion, in order to show $h \in R_\alpha[a, b]$ it is enough to show that

$$\forall \epsilon > 0 \exists P \in \Pi \text{ such that } U(h, \alpha, P) - L(h, \alpha, P) < \epsilon.$$

Let $\epsilon > 0$ be given. Our goal is to find $P \in \Pi$ such that

$$U(h, \alpha, P) - L(h, \alpha, P) < \epsilon.$$

Let $\tilde{M} = \sup_{a \leq x \leq b} |h(x)|$. Let $\hat{\epsilon} = \frac{\epsilon}{[\alpha(b) - \alpha(a)] + 2\tilde{M} + 1}$. We have

$$\left. \begin{array}{l} \varphi \text{ is continuous on } [m, M] \\ [m, M] \text{ is compact} \end{array} \right\} \implies \varphi \text{ is uniformly continuous on } [m, M].$$

so

$$\exists 0 < \delta < \hat{\epsilon} \text{ such that } \forall s, t \in [m, M] \text{ if } |s - t| < \delta \text{ then } |\varphi(s) - \varphi(t)| < \hat{\epsilon}$$

Also,

$$f \in R_\alpha[a, b] \implies \exists \tilde{P} \text{ such that } U(f, \alpha, \tilde{P}) - L(f, \alpha, \tilde{P}) = \sum_{k=1}^n (M_k - m_k) \Delta\alpha_k < \delta^2$$

We claim that this $\tilde{P}$ can be used as the P we are looking for. Indeed, let

$$\forall 1 \leq k \leq n \quad m_k^* = \inf \{h(x) : x \in [x_{k-1}, x_k]\}, \quad M_k^* = \sup \{h(x) : x \in [x_{k-1}, x_k]\}.$$

Note that

$$U(h, \alpha, \tilde{P}) - L(h, \alpha, \tilde{P}) = \sum_{k=1}^n (M_k^* - m_k^*) \Delta\alpha_k$$

In what follows, we will show that the above sum is less than ϵ . Divide the numbers $1, \dots, n$ into two cases:

$$A = \{k : M_k - m_k < \delta\} \quad B = \{k : M_k - m_k \geq \delta\}$$

We have

$$U(h, \alpha, \tilde{P}) - L(h, \alpha, \tilde{P}) = \sum_{k=1}^n (M_k^* - m_k^*) \Delta\alpha_k = \sum_{k \in A} (M_k^* - m_k^*) \Delta\alpha_k + \sum_{k \in B} (M_k^* - m_k^*) \Delta\alpha_k \quad (\text{I})$$

*) If $k \in A$:

$$\begin{aligned} M_k - m_k < \delta &\implies \sup_{x \in [x_{k-1}, x_k]} |f(x) - f(y)| < \delta \\ &\implies \forall x, y \in [x_{k-1}, x_k] \quad |f(x) - f(y)| < \delta \\ &\implies \forall x, y \in [x_{k-1}, x_k] \quad |\varphi(f(x)) - \varphi(f(y))| < \hat{\epsilon} \\ &\implies \forall x, y \in [x_{k-1}, x_k] \quad |h(x) - h(y)| < \hat{\epsilon} \\ &\implies \sup_{x, y \in [x_{k-1}, x_k]} |h(x) - h(y)| \leq \hat{\epsilon} \\ &\implies M_k^* - m_k^* \leq \hat{\epsilon} \end{aligned} \quad (\text{II})$$

*) If $k \in B$, then

$$\begin{aligned} \delta \sum_{k \in B} \Delta\alpha_k &= \sum_{k \in B} \delta \Delta\alpha_k \leq \sum_{k \in B} (M_k - m_k) \Delta\alpha_k \leq \sum_{k=1}^n (M_k - m_k) \Delta\alpha_k = U(f, \alpha, \tilde{P}) - L(f, \alpha, \tilde{P}) < \delta^2 \\ &\implies \delta \sum_{k \in B} \Delta\alpha_k < \delta^2 \implies \sum_{k \in B} \Delta\alpha_k < \delta \implies \sum_{k \in B} \Delta\alpha_k < \hat{\epsilon} \end{aligned} \quad (\text{III})$$

It follows from (I), (II), and (III) that

$$\begin{aligned}
 \sum_{k=1}^n (M_k^* - m_k^*) \Delta\alpha_k &= \sum_{k \in A} (M_k^* - m_k^*) \Delta\alpha_k + \sum_{k \in B} (M_k^* - m_k^*) \Delta\alpha_k \\
 &\leq \sum_{k \in A} \hat{\epsilon} \Delta\alpha_k + \sum_{k \in B} 2\tilde{M} \Delta\alpha_k \\
 &= \hat{\epsilon} \sum_{k \in A} \Delta\alpha_k + 2\tilde{M} \sum_{k \in B} \Delta\alpha_k \\
 &\leq \hat{\epsilon} \sum_{k=1}^n \Delta\alpha_k + 2\tilde{M} \hat{\epsilon} \\
 &= \hat{\epsilon} [\alpha(b) - \alpha(a) + 2\tilde{M}] \\
 &= \frac{\epsilon}{\alpha(b) - \alpha(a) + 2\tilde{M} + 1} [\alpha(b) - \alpha(a) + 2\tilde{M}] \\
 &< \epsilon
 \end{aligned}$$

□

Example 2.2.3. If $f \in R_\alpha[a, b]$, then $|f| \in R_\alpha[a, b]$.
 If $f \in R_\alpha[a, b]$, then $f^2 \in R_\alpha[a, b]$.

2.3 The Sequential Criterion of Integrability

Theorem 2.3.1. (The Sequential Criterion of Integrability)

Suppose $f : [a, b] \rightarrow \mathbb{R}$ is bounded and $\alpha : [a, b] \rightarrow \mathbb{R}$ is increasing.

- (i) If $f \in R_\alpha[a, b]$, then there exists a sequence of partitions $(P_n)_{n \geq 1}$ in $\Pi[a, b]$ such that

$$\lim_{n \rightarrow \infty} [U(f, \alpha, P_n) - L(f, \alpha, P_n)] = 0.$$

- (ii) If there exists a sequence of partitions $(P_n)_{n \geq 1}$ in $\Pi[a, b]$ such that

$$\lim_{n \rightarrow \infty} [U(f, \alpha, P_n) - L(f, \alpha, P_n)] = 0,$$

then $f \in R_\alpha[a, b]$ and

$$\int_a^b f d\alpha = \lim_{n \rightarrow \infty} U(f, \alpha, P_n) = \lim_{n \rightarrow \infty} L(f, \alpha, P_n).$$

Proof. (i) $f \in R_\alpha[a, b] \iff \forall \epsilon > 0, \exists P_\epsilon \in \Pi$ such that $U(f, \alpha, P_\epsilon) - L(f, \alpha, P_\epsilon) < \epsilon$. In particular,

$$\begin{aligned} \epsilon = 1 &\implies \exists P_1 \in \Pi \text{ such that } U(f, \alpha, P_1) - L(f, \alpha, P_1) < 1 \\ \epsilon = \frac{1}{2} &\implies \exists P_2 \in \Pi \text{ such that } U(f, \alpha, P_2) - L(f, \alpha, P_2) < \frac{1}{2} \\ \epsilon = \frac{1}{3} &\implies \exists P_3 \in \Pi \text{ such that } U(f, \alpha, P_3) - L(f, \alpha, P_3) < \frac{1}{3} \\ &\vdots \end{aligned}$$

In this way, we obtain a sequence $(P_n)_{n \geq 1}$ of partitions such that

$$\forall n \geq 1 \quad U(f, \alpha, P_n) - L(f, \alpha, P_n) < \frac{1}{n}.$$

It follows from the squeeze theorem that

$$\lim_{n \rightarrow \infty} [U(f, \alpha, P_n) - L(f, \alpha, P_n)] = 0.$$

- (ii) According the Cauchy Criterion, it is enough to show that

$$\forall \epsilon > 0 \quad \exists P \in \Pi[a, b] \text{ such that } U(f, \alpha, P) - L(f, \alpha, P) < \epsilon.$$

Let $\epsilon > 0$ be given. Our goal is to find $P \in \Pi$ such that

$$U(f, \alpha, P) - L(f, \alpha, P) < \epsilon.$$

Since

$$\lim_{n \rightarrow \infty} [U(f, \alpha, P_n) - L(f, \alpha, P_n)] = 0,$$

there exists $N \in \mathbb{N}$ such that

$$\forall n \in \mathbb{N} \quad U(f, \alpha, P_n) - L(f, \alpha, P_n) < \epsilon.$$

In particular, P_{N+1} can be used as the P we were looking for. It remains to show that $\int_a^b f d\alpha = \lim_{n \rightarrow \infty} U(f, \alpha, P_n) = \lim_{n \rightarrow \infty} L(f, \alpha, P_n)$. We have

$$0 \leq U(f, \alpha, P_n) - U(f, \alpha) \leq U(f, \alpha, P_n) - L(f, \alpha, P_n) \leq U(f, \alpha, P_n) - L(f, \alpha, P_n) \rightarrow 0 \text{ as } n \rightarrow \infty.$$

It follows from the squeeze theorem that

$$\lim_{n \rightarrow \infty} [U(f, \alpha, P_n) - U(f, \alpha)] = 0.$$

Thus

$$\lim_{n \rightarrow \infty} U(f, \alpha, P_n) = U(f, \alpha) = \int_a^b f d\alpha.$$

Similary,

$$\begin{aligned}
 0 \leq L(f, \alpha) - L(f, \alpha, P_n) &\leq U(f, \alpha) - L(f, \alpha, P_n) \leq U(f, \alpha, P_n) - L(f, \alpha, P_n) \rightarrow 0 \text{ as } n \rightarrow \infty \\
 &\stackrel{\text{SQZ Thm}}{\implies} \lim_{n \rightarrow \infty} [L(f, \alpha) - L(f, \alpha, P_n)] \\
 &\implies \lim_{n \rightarrow \infty} L(f, \alpha, P_n) = L(f, \alpha) = \int_a^b f d\alpha.
 \end{aligned}$$

□

Theorem 2.3.2. (Algebraic Limit Theorem for Riemann-Stieltjes Integrals)

Suppose $f, g \in R_\alpha[a, b]$. Then

- (i) $\forall k \in \mathbb{R} \quad kf \in R_\alpha[a, b]$ with $\int_a^b kf d\alpha = k \int_a^b f d\alpha$.
- (ii) $f + g \in R_\alpha[a, b]$ with $\int_a^b f + g d\alpha = \int_a^b f d\alpha + \int_a^b g d\alpha$.
- (iii) $f^2 \in R_\alpha[a, b]$, and $fg \in R_\alpha[a, b]$.
- (iv) If $g \neq 0$ on $[a, b]$ and $\frac{1}{g}$ is bounded on $[a, b]$, then $\frac{1}{g} \in R_\alpha[a, b]$ and $\frac{f}{g} \in R_\alpha[a, b]$.

Remark 2.3.1. (1) As a result of (i) and (ii), the set $R_\alpha[a, b]$ is a vector space. Moreover, the function

$$\begin{aligned}
 T : R_\alpha[a, b] &\rightarrow \mathbb{R} \\
 T(f) &= \int_a^b f d\alpha
 \end{aligned}$$

is linear.

- (2) The function $g : [0, 1] \rightarrow \mathbb{R}$ defined by

$$g(x) = \begin{cases} x & 0 < x \leq 1 \\ 1 & x = 0 \end{cases}$$

is an example of a bounded function which is never zero, but $\frac{1}{g}$ is unbounded.

Notation . Let $h : A \subseteq \mathbb{R} \rightarrow \mathbb{R}$ be a bounded function.

- (i) We denote $\sup \{h(x) : x \in A\}$ by $\sup_A h$ or $\sup_A h(x)$ or $\sup_{x \in A} h(x)$.
- (ii) We denote $\inf \{h(x) : x \in A\}$ by $\inf_A h$ or $\inf_A h(x)$ or $\inf_{x \in A} h(x)$.

Lemma 2.3.1. Let $A \subseteq \mathbb{R}$. Let $f : A \rightarrow \mathbb{R}$ and $g : A \rightarrow \mathbb{R}$ be bounded functions. Then

- (i) $\sup_A (f + g) \leq \sup_A f + \sup_A g$.
- (ii) $\inf_A (f + g) \geq \inf_A f + \inf_A g$.
- (iii) $\forall k \geq 0 \quad \sup_A (kf) = k \sup_A f$ and $\inf_A (kf) = k \inf_A f$.
- (iv) $\forall k \leq 0 \quad \sup_A (kf) = k \inf_A f$ and $\inf_A (kf) = k \sup_A f$.
- (v) $\sup \{|f(x) - f(y)| : x, y \in A\} = \sup_A f - \inf_A f$.
- (vi) If $\exists k > 0$ such that $\forall z, w \in A \quad |f(z) - f(w)| \leq k|g(z) - g(w)|$, then

$$\sup_A f - \inf_A f \leq k \left[\sup_A g - \inf_A g \right]$$

Lemma 2.3.2. Let $f : [a, b] \rightarrow \mathbb{R}$ and $g : [a, b] \rightarrow \mathbb{R}$ be bounded, $\alpha : [a, b] \rightarrow \mathbb{R}$ be increasing, and $P \in \Pi[a, b]$. Then

- (i) $U(f + g, \alpha, P) \leq U(f, \alpha, P) + U(g, \alpha, P)$.

- (ii) $L(f + g, \alpha, P) \geq L(f, \alpha, P) + L(g, \alpha, P)$.
- (iii) $\forall k \geq 0 \quad U(kf, \alpha, P) = kU(f, \alpha, P)$ and $L(kf, \alpha, P) = kL(f, \alpha, P)$.
- (iv) $\forall k < 0 \quad U(kf, \alpha, P) = kL(f, \alpha, P)$ and $L(kf, \alpha, P) = kU(f, \alpha, P)$.

Proof. Proof of the Algebraic Limit Theorem for Integrals.

- (i) $\forall k \in \mathbb{R}, \quad kf \in R_\alpha[a, b]$ with $\int_a^b kf d\alpha = k \int_a^b f d\alpha$:

Note that

$$f \in R_\alpha[a, b] \implies f \text{ is bounded on } [a, b] \implies kf \text{ is bounded on } [a, b].$$

Since $f \in R_\alpha[a, b]$, there exists a sequence $(P_n)_{n \geq 1}$ in $\Pi[a, b]$ such that

$$\lim_{n \rightarrow \infty} [U(f, \alpha, P_n) - L(f, \alpha, P_n)] = 0, \quad \int_a^b f d\alpha = \lim_{n \rightarrow \infty} U(f, \alpha, P_n) = \lim_{n \rightarrow \infty} L(f, \alpha, P_n).$$

In what follows, we'll show that

$$\lim_{n \rightarrow \infty} [U(kf, \alpha, P_n) - L(kf, \alpha, P_n)] = 0$$

and so $kf \in R_\alpha[a, b]$. We may consider two cases:

Case 1: $k \geq 0$

$$\begin{aligned} \lim_{n \rightarrow \infty} [U(kf, \alpha, P_n) - L(kf, \alpha, P_n)] &= \lim_{n \rightarrow \infty} [kU(f, \alpha, P_n) - kL(f, \alpha, P_n)] \\ &= k \lim_{n \rightarrow \infty} [U(f, \alpha, P_n) - L(f, \alpha, P_n)] \\ &= 0. \end{aligned}$$

Case 2: $k < 0$

$$\begin{aligned} \lim_{n \rightarrow \infty} [U(kf, \alpha, P_n) - L(kf, \alpha, P_n)] &= \lim_{n \rightarrow \infty} [kL(f, \alpha, P_n) - kU(f, \alpha, P_n)] \\ &= -k \lim_{n \rightarrow \infty} [U(f, \alpha, P_n) - L(f, \alpha, P_n)] \\ &= 0. \end{aligned}$$

Moreover, if $k \geq 0$, then

$$\int_a^b kf d\alpha = \lim_{n \rightarrow \infty} U(kf, \alpha, P_n) = \lim_{n \rightarrow \infty} k \cdot U(f, \alpha, P_n) = k \cdot \int_a^b f d\alpha.$$

Similarly, for $k < 0$,

$$\int_a^b kf d\alpha = \lim_{n \rightarrow \infty} U(kf, \alpha, P_n) = \lim_{n \rightarrow \infty} k \cdot L(f, \alpha, P_n) = k \cdot \int_a^b f d\alpha.$$

- (ii) $f + g \in R_\alpha[a, b]$ with $\int_a^b f + g d\alpha = \int_a^b f d\alpha + \int_a^b g d\alpha$:

Note that

$$\left. \begin{aligned} f \in R_\alpha[a, b] &\implies f \text{ is bounded on } [a, b] \\ g \in R_\alpha[a, b] &\implies g \text{ is bounded on } [a, b] \end{aligned} \right\} \implies f + g \text{ is bounded on } [a, b].$$

Since $f \in R_\alpha[a, b]$ there exists $(P_n)_{n \geq 1}$ in $\Pi[a, b]$ such that

$$\lim_{n \rightarrow \infty} [U(f, \alpha, P_n) - L(f, \alpha, P_n)] = 0, \quad \int_a^b f d\alpha = \lim_{n \rightarrow \infty} U(f, \alpha, P_n) = \lim_{n \rightarrow \infty} L(f, \alpha, P_n).$$

Since $g \in R_\alpha[a, b]$ there exists $(Q_n)_{n \geq 1}$ in $\Pi[a, b]$ such that

$$\lim_{n \rightarrow \infty} [U(g, \alpha, Q_n) - L(g, \alpha, Q_n)] = 0, \quad \int_a^b g d\alpha = \lim_{n \rightarrow \infty} U(g, \alpha, Q_n) = \lim_{n \rightarrow \infty} L(g, \alpha, Q_n).$$

In what follows, we will show that

$$\lim_{n \rightarrow \infty} [U(f + g, \alpha, P_n \cup Q_n) - L(f + g, \alpha, P_n \cup Q_n)] = 0$$

and so $f + g \in R_\alpha[a, b]$. We have

$$\begin{aligned} L(f + g, \alpha, P_n \cup Q_n) &\geq L(f, \alpha, P_n \cup Q_n) + L(g, \alpha, P_n \cup Q_n) \\ &\geq L(f, \alpha, P_n) + L(g, \alpha, Q_n) \end{aligned}$$

and

$$\begin{aligned} U(f + g, \alpha, P_n \cup Q_n) &\leq U(f, \alpha, P_n \cup Q_n) + U(g, \alpha, P_n \cup Q_n) \\ &\leq U(f, \alpha, P_n) + U(g, \alpha, Q_n) \end{aligned}$$

so

$$\begin{aligned} L(f, \alpha, P_n) + L(g, \alpha, Q_n) &\leq L(f + g, \alpha, P_n \cup Q_n) \\ &\leq U(f + g, \alpha, P_n \cup Q_n) \\ &\leq U(f, \alpha, P_n) + U(g, \alpha, Q_n). \end{aligned}$$

Therefore,

$$\begin{aligned} 0 &\leq U(f + g, \alpha, P_n \cup Q_n) - L(f + g, \alpha, P_n \cup Q_n) \\ &\leq [U(f, \alpha, P_n) - L(f, \alpha, P_n)] + [U(g, \alpha, Q_n) - L(g, \alpha, Q_n)] \rightarrow 0 + 0 = 0. \end{aligned}$$

By the squeeze theorem,

$$\lim_{n \rightarrow \infty} [U(f + g, \alpha, P_n \cup Q_n) - L(f + g, \alpha, P_n \cup Q_n)] = 0.$$

Moreover,

$$\begin{aligned} \int_a^b f + g d\alpha &= \lim_{n \rightarrow \infty} [U(f + g, \alpha, P_n \cup Q_n)] \\ &\leq \lim_{n \rightarrow \infty} [U(f, \alpha, P_n) + U(g, \alpha, P_n)] = \int_a^b f d\alpha + \int_a^b g d\alpha \\ \int_a^b f + g d\alpha &= \lim_{n \rightarrow \infty} [L(f + g, \alpha, P_n \cup Q_n)] \\ &\geq \lim_{n \rightarrow \infty} [L(f, \alpha, P_n) + L(g, \alpha, P_n)] = \int_a^b f d\alpha + \int_a^b g d\alpha. \end{aligned}$$

This implies $\int_a^b f + g d\alpha = \int_a^b f d\alpha + \int_a^b g d\alpha$.

(iii) $f^2 \in R_\alpha[a, b]$:

Note that $\phi : \mathbb{R} \rightarrow \mathbb{R}$, $\phi(x) = x^2$ is continuous, so since $f \in R_\alpha[a, b]$, $\phi \circ f \in R_\alpha[a, b] \implies f^2 \in R_\alpha[a, b]$.

$fg \in R_\alpha[a, b]$:

Note that

$$\left. \begin{aligned} (f + g)^2 &= f^2 + 2fg + g^2 \\ (f - g)^2 &= f^2 - 2fg + g^2 \end{aligned} \right\} \implies 4fg = (f + g)^2 - (f - g)^2 \implies fg = \frac{1}{4} [(f + g)^2 - (f - g)^2].$$

Therefore,

$$\begin{aligned} f, g \in R_\alpha[a, b] &\stackrel{(i)}{\implies} f, g, -g \in R_\alpha[a, b] \\ &\stackrel{(ii)}{\implies} f + g, f - g \in R_\alpha[a, b] \\ &\stackrel{(iii)}{\implies} (f + g)^2, (f - g)^2 \in R_\alpha[a, b] \\ &\stackrel{(i)}{\implies} (f + g)^2, -(f - g)^2 \in R_\alpha[a, b] \\ &\stackrel{(ii)}{\implies} (f + g)^2 - (f - g)^2 \in R_\alpha[a, b] \\ &\stackrel{(i)}{\implies} \frac{1}{4} [(f + g)^2 - (f - g)^2] \in R_\alpha[a, b] \\ &\implies fg \in R_\alpha[a, b]. \end{aligned}$$

□

Theorem 2.3.3. (Order Properties of Riemann-Stieltjes Integrals)

Suppose $f, g \in R_\alpha[a, b]$. Then

- (i) If $m \leq f(x) \leq M \quad \forall x \in [a, b]$, then $m(\alpha(b) - \alpha(a)) \leq \int_a^b f d\alpha \leq M(\alpha(b) - \alpha(a))$.
- (ii) If $f \leq g$ on $[a, b]$, then $\int_a^b f d\alpha \leq \int_a^b g d\alpha$.

Proof. (i) Note that for any $P \in \Pi[a, b]$,

$$\begin{aligned} \int_a^b f d\alpha &= L(f, \alpha) \geq L(f, \alpha, P) \\ \int_a^b f d\alpha &= U(f, \alpha) \leq U(f, \alpha, P). \end{aligned}$$

In particular, for the partition $P = \{a, b\}$, we have

$$\begin{aligned} \int_a^b f d\alpha &\geq L(f, \alpha, P) = \inf \{f(x) : x \in [a, b]\} (\alpha(b) - \alpha(a)) \geq m(\alpha(b) - \alpha(a)) \\ \int_a^b f d\alpha &\leq U(f, \alpha, P) = \sup \{f(x) : x \in [a, b]\} (\alpha(b) - \alpha(a)) \leq M(\alpha(b) - \alpha(a)) \end{aligned}$$

- (ii) Let $h = g - f$. We have $h \geq 0$, so by part (i)

$$\begin{aligned} 0(\alpha(b) - \alpha(a)) &\leq \int_a^b h d\alpha \\ &= \int_a^b g d\alpha - \int_a^b f d\alpha \\ \implies \int_a^b f d\alpha &\leq \int_a^b g d\alpha. \end{aligned}$$

□

Theorem 2.3.4. ($|\int_a^b f d\alpha| \leq \int_a^b |f| d\alpha$)

Suppose $f \in R_\alpha[a, b]$. Then

- (i) $|f| \in R_\alpha[a, b]$.
- (ii) $|\int_a^b f d\alpha| \leq \int_a^b |f| d\alpha$.

Proof. (i) Since $\phi : \mathbb{R} \rightarrow \mathbb{R}$, $\phi(x) = |x|$ is continuous and f is Riemann-Stieltjes integrable, it follows that $|f| = \phi \circ f$ is Riemann-Stieltjes integrable. Hence $|f| \in R_\alpha[a, b]$.

- (ii) Recall that $|t| \leq s \iff -s \leq t \leq s$. So our goal is to show that

$$-\int_a^b |f| d\alpha \leq \int_a^b f d\alpha \leq \int_a^b |f| d\alpha$$

Clearly, (for any c , $-|c| \leq c \leq |c|$)

$$\begin{aligned} -|f(x)| &\leq f(x) \leq |f(x)| \quad \forall x \in [a, b] \\ \implies -\int_a^b |f| d\alpha &\leq \int_a^b f d\alpha \leq \int_a^b |f| d\alpha \quad \forall x \in [a, b] \end{aligned}$$

□

Theorem 2.3.5. (The Mean Value Theorem for Riemann-Stieltjes Integrals)

Suppose $f : [a, b] \rightarrow \mathbb{R}$ is continuous, $\alpha : [a, b] \rightarrow \mathbb{R}$ is increasing, and $\alpha(a) \neq \alpha(b)$. Then

$$\exists c \in [a, b] \text{ such that } f(c) = \frac{\int_a^b f d\alpha}{\alpha(b) - \alpha(a)}$$

Proof. Because f is continuous on $[a, b]$, and $[a, b]$ is compact, by the Extreme Value Theorem f attains its maximum and minimum on $[a, b]$. Let $m = \min \{f(x) : x \in [a, b]\}$ and $M = \max \{f(x) : x \in [a, b]\}$. We have

$$\begin{aligned} \forall x \in [a, b] \quad m &\leq f(x) \leq M \\ \implies m(\alpha(b) - \alpha(a)) &\leq \int_a^b f d\alpha \leq M(\alpha(b) - \alpha(a)) \\ \implies m &\leq \frac{\int_a^b f d\alpha}{\alpha(b) - \alpha(a)} \leq M. \end{aligned}$$

Since f is continuous, it follows from the Intermediate Value Theorem

$$\exists c \in [a, b] \text{ such that } f(c) = \frac{\int_a^b f d\alpha}{\alpha(b) - \alpha(a)}$$

□

Theorem 2.3.6. (Generalized Mean Value Theorem for Riemann-Stieltjes Integrals)

Suppose $f : [a, b] \rightarrow \mathbb{R}$ is continuous, $\alpha : [a, b] \rightarrow \mathbb{R}$ is increasing, $g \in R_\alpha[a, b]$, and either $g \geq 0$ on $[a, b]$ or $g \leq 0$ on $[a, b]$. Then

$$\exists c \in [a, b] \text{ such that } \int_a^b f g d\alpha = f(c) \int_a^b g d\alpha.$$

Proof. We will prove the claim for the case $g \geq 0$ on $[a, b]$. The proof for the case $g \leq 0$ is analogous. Since f is continuous on $[a, b]$ and $[a, b]$ is compact, f contains its maximum and minimum on $[a, b]$. Let $M = \max \{f(x) : x \in [a, b]\}$ and $m = \min \{f(x) : x \in [a, b]\}$. Then, since $g \geq 0$ on $[a, b]$,

$$\begin{aligned} \forall x \in [a, b] \quad m &\leq f(x) \leq M \\ \implies \forall x \in [a, b] \quad m g(x) &\leq f(x) g(x) \leq M g(x) \\ \implies \forall x \in [a, b] \quad m \int_a^b g d\alpha &\leq \int_a^b f g d\alpha \leq M \int_a^b g d\alpha. \end{aligned}$$

Now, we may consider two cases:

Case 1: $\int_a^b g d\alpha = 0$ If $\int_a^b g d\alpha = 0$, it follows that $0 \leq \int_a^b f g d\alpha \leq 0$ and so $\int_a^b f g d\alpha = 0$. So in this case, the claim holds for every $c \in [a, b]$:

$$\int_a^b f g d\alpha = f(c) \int_a^b g d\alpha.$$

Case 2: $\int_a^b g d\alpha \neq 0$ In this case, since $g \geq 0$, it follows that $\int_a^b g d\alpha > 0$ and

$$m \leq \frac{\int_a^b f g d\alpha}{\int_a^b g d\alpha} \leq M.$$

Since f is continuous, it follows from the Intermediate Value Theorem that

$$\exists c \in [a, b] \text{ such that } f(c) = \frac{\int_a^b f g d\alpha}{\int_a^b g d\alpha}.$$

That is, $\int_a^b f g d\alpha = f(c) \int_a^b g d\alpha$.

□

Theorem 2.3.7. (Additivity for Riemann-Stieltjes Integrals)

Suppose $f : [a, b] \rightarrow \mathbb{R}$ is continuous, $\alpha : [a, b] \rightarrow \mathbb{R}$ is increasing, and $c \in (a, b)$. Then

$$f \in R_\alpha[a, b] \iff f \in R_\alpha[a, c] \text{ and } f \in R_\alpha[c, b].$$

In this case, we have

$$\int_a^b f d\alpha = \int_a^c f d\alpha + \int_c^b f d\alpha.$$



2.4 The Fundamental Theorem of Calculus and Techniques of Integration

Lemma 2.4.1.

Suppose $f : [a, b] \rightarrow \mathbb{R}$ is bounded, $\alpha : [a, b] \rightarrow \mathbb{R}$ is increasing, $P = \{x_0, \dots, x_n\} \in \Pi[a, b]$, and $R \in \mathbb{R}$.

- (i) If for all tags $(s_k)_{a \leq k \leq n}$ of P we have $\sum_{k=1}^n f(s_k) \Delta \alpha_k \leq R$, then $U(f, \alpha, P) \leq R$.
- (ii) If for all tags $(s_k)_{a \leq k \leq n}$ of P we have $R \leq \sum_{k=1}^n f(s_k) \Delta \alpha_k$, then $L(f, \alpha, P) \geq R$.

Proof. (i) If α is constant, then $\sum_{k=1}^n f(s_k) \Delta \alpha_k = 0$ and $U(f, \alpha, P) = \sum_{k=1}^n M_k \Delta \alpha_k = 0$, so the claim obviously holds. Suppose $\alpha(b) \neq \alpha(a)$. It's enough to show

$$\forall \epsilon > 0 \quad U(f, \alpha, P) \leq R + \epsilon.$$

Let $\epsilon > 0$ be given. For each $k \in \{1, \dots, n\}$ we have

$$M_k = \sup \{f(x) : x \in [x_{k-1}, x_k]\} \implies \exists s_k \in [x_{k-1}, x_k] \text{ such that } M_k - \frac{\epsilon}{\alpha(b) - \alpha(a)} < f(s_k).$$

We have

$$\begin{aligned} U(f, \alpha, P) &= \sum_{k=1}^n M_k \Delta \alpha_k < \sum_{k=1}^n \left[f(s_k) + \frac{\epsilon}{\alpha(b) - \alpha(a)} \right] \Delta \alpha_k \\ &= \sum_{k=1}^n f(s_k) \Delta \alpha_k + \frac{\epsilon}{\alpha(b) - \alpha(a)} \sum_{k=1}^n \Delta \alpha_k \\ &\leq R + \epsilon \end{aligned}$$

as desired.

- (ii) Completely analogous to (i). □

Theorem 2.4.1.

Suppose $f : [a, b] \rightarrow \mathbb{R}$ is bounded, $\alpha : [a, b] \rightarrow \mathbb{R}$ is increasing, and $\alpha' \in R[a, b]$. Then

$$f \in R_\alpha[a, b] \iff f\alpha' \in R[a, b]$$

and in this case,

$$\int_a^b f d\alpha = \int_a^b f(x) \alpha'(x) dx.$$

Proof. It is enough to show that

$$\begin{aligned} U(f, \alpha) &= U(f\alpha') \\ L(f, \alpha) &= L(f\alpha') \end{aligned} \tag{*}$$

Indeed, if we prove (*), then we can write

$$\begin{aligned} f \in R_\alpha[a, b] &\iff U(f, \alpha) = L(f, \alpha) \\ &\iff U(f\alpha') = L(f\alpha') \\ &\iff f\alpha' \in R[a, b]. \end{aligned}$$

Moreover,

$$\int_a^b f d\alpha = U(f, \alpha) \stackrel{(*)}{=} U(f\alpha') = \int_a^b f(x) \alpha'(x) dx.$$

In what follows, we will prove $U(f, \alpha) = U(f\alpha')$. The proof for $L(f, \alpha) = L(f\alpha')$ is completely analogous.

let $P = \{x_0, \dots, x_n\}$ be any partition of $[a, b]$. Let (s_k) be any tag of P . We have

$$\begin{aligned}
 \left| \sum_{k=1}^n f(s_k) \Delta \alpha_k - \sum_{k=1}^n f(s_k) \alpha'(s_k) \Delta x_k \right| &= \left| \sum_{k=1}^n f(s_k) \alpha'(t_k) \Delta x_k - \sum_{k=1}^n f(s_k) \alpha'(s_k) \Delta x_k \right| \quad (\text{MVT on } \Delta \alpha_k) \\
 &= \left| \sum_{k=1}^n f(s_k) (\alpha'(t_k) - \alpha'(s_k)) \Delta x_k \right| \\
 &\leq \sum_{k=1}^n |f(s_k)| \cdot |\alpha'(t_k) - \alpha'(s_k)| \Delta x_k \\
 &\leq \hat{M} \sum_{k=1}^n |\alpha'(t_k) - \alpha'(s_k)| \Delta x_k \quad (\hat{M} = \sup_{a \leq x \leq b} |f(x)|) \\
 &\leq \hat{M} \sum_{k=1}^n \left[\sup_{I_k} \alpha' - \inf_{I_k} \alpha' \right] \Delta x_k \quad (\text{Lemma 2.2.1}) \\
 &= \hat{M} [U(\alpha', P) - L(\alpha', P)]
 \end{aligned}$$

So,

$$\left| \sum_{k=1}^n f(s_k) \Delta \alpha_k - \sum_{k=1}^n f(s_k) \alpha'(s_k) \Delta x_k \right| \leq \hat{M} [U(\alpha', P) - L(\alpha', P)].$$

Therefore,

$$\begin{aligned}
 (1) \quad \sum_{k=1}^n f(s_k) \Delta \alpha_k &\leq \sum_{k=1}^n f(s_k) \alpha'(s_k) \Delta x_k + \hat{M} [U(\alpha', P) - L(\alpha', P)] \\
 &\leq U(f\alpha', P) + \hat{M} [U(\alpha', P) - L(\alpha', P)].
 \end{aligned}$$

So, by Lemma 2.4.1,

$$U(f, \alpha, P) \leq U(f\alpha', P) + \hat{M} [U(\alpha', P) - L(\alpha', P)].$$

$$(2) \quad \sum_{k=1}^n f(s_k) \alpha'(s_k) \Delta x_k \leq \sum_{k=1}^n f(s_k) \Delta \alpha_k + \hat{M} [U(\alpha', P) - L(\alpha', P)]$$

So, by Lemma 2.4.1,

$$U(f\alpha', P) \leq U(f, \alpha, P) + \hat{M} [U(\alpha', P) - L(\alpha', P)].$$

It follows from (1) and (2) that

$$(3) \quad |U(f, \alpha, P) - U(f\alpha', P)| \leq \hat{M} [U(\alpha', P) - L(\alpha', P)].$$

Note that

$$\begin{aligned}
 U(f, \alpha) &= \inf \{U(f, \alpha, P) : P \in \Pi\} \implies \exists (P_n^{(1)}) \subseteq \Pi \text{ such that } \lim_{n \rightarrow \infty} U(f, \alpha, P_n^{(1)}) = U(f, \alpha) \\
 U(f\alpha') &= \inf \{U(f\alpha', P) : P \in \Pi\} \implies \exists (P_n^{(2)}) \subseteq \Pi \text{ such that } \lim_{n \rightarrow \infty} U(f\alpha', P_n^{(2)}) = U(f\alpha') \\
 \alpha' \in R[a, b] &\implies \exists (P_n^{(3)}) \subseteq \Pi \text{ such that } \lim_{n \rightarrow \infty} [U(\alpha', P_n^{(3)}) - L(\alpha', P_n^{(3)})] = 0.
 \end{aligned}$$

For each $n \in \mathbb{N}$, let $P_n = P_n^{(1)} \cup P_n^{(2)} \cup P_n^{(3)}$. By the Squeeze Theorem,

$$\begin{aligned}
 \forall n \geq 1 \quad U(f, \alpha) &\leq U(f, \alpha, P_n) \leq U(f, \alpha, P_n^{(1)}) \implies \lim_{n \rightarrow \infty} U(f, \alpha, P_n) = U(f, \alpha) \\
 \forall n \geq 1 \quad U(f\alpha') &\leq U(f\alpha', P_n) \leq U(f\alpha', P_n^{(2)}) \implies \lim_{n \rightarrow \infty} U(f\alpha', P_n) = U(f\alpha').
 \end{aligned}$$

Since P_n is a refinement of $P_n^{(3)}$,

$$0 \leq [U(\alpha', P_n) - L(\alpha', P_n)] \leq U(\alpha', P_n^{(3)}) - L(\alpha', P_n^{(3)}) \implies \lim_{n \rightarrow \infty} [U(\alpha', P_n) - L(\alpha', P_n)] = 0.$$

It follows from (3) that

$$\forall n \geq 1 \quad 0 \leq |U(f, \alpha, P_n) - U(f\alpha', P_n)| \leq \hat{M} [U(\alpha', P_n) - L(\alpha', P_n)].$$

So, by the Squeeze Theorem,

$$\begin{aligned}
 & \lim_{n \rightarrow \infty} |U(f, \alpha, P_n) - U(f\alpha', P_n)| = 0 \\
 \implies & \left| \lim_{n \rightarrow \infty} (U(f, \alpha, P_n) - U(f\alpha', P_n)) \right| = 0 \\
 \implies & |U(f, \alpha) - U(f\alpha')| = 0 \\
 \implies & U(f, \alpha) - U(f\alpha') = 0 \\
 \implies & U(f, \alpha) = U(f\alpha')
 \end{aligned}$$

□

Theorem 2.4.2. (Change of Variables for Integrals)

Let $f \in R_\alpha[a, b]$, and let $\phi : [A, B] \rightarrow [a, b]$ be onto and strictly increasing. If we let $g = f \circ \phi$ and $\beta = \alpha \circ \phi$, then

$$g \in R_\beta[A, B] \text{ and } \int_a^b f d\alpha = \int_A^B g d\beta.$$

Remark 2.4.1. Consider the special case where $\alpha(x) = x$ and $\phi' \in \mathbb{R}$. In this case, $\beta = \alpha \circ \phi = \phi$. So, the conclusion of the theorem can be written as

$$\int_a^b f(x) dx = \int_A^B (f \circ \phi) d\phi = \int_A^B f(\phi(y)) \phi'(y) dy.$$

Proof. First, note that

$$\left. \begin{array}{l} \alpha : [a, b] \rightarrow \mathbb{R} \text{ is increasing} \\ \phi : [A, B] \rightarrow \mathbb{R} \text{ is increasing} \end{array} \right\} \implies \beta = \alpha \circ \phi \text{ is increasing on } [A, B]$$

and

$$\left. \begin{array}{l} f \in R_\alpha[a, b] \implies f \text{ is bounded on } [a, b] \\ \forall x \in [A, B] \ a \leq \phi(x) \leq b \implies \phi \text{ is bounded on } [A, B] \end{array} \right\} \implies g = f \circ \phi \text{ is bounded on } [A, B].$$

Also, note that there is a 1-1 correspondance between $\Pi[a, b]$ and $\Pi[A, B]$:

$$\begin{aligned}
 H : \Pi[a, b] &\rightarrow \Pi[A, B] \\
 P = \{x_0, \dots, x_n\} &\mapsto Q = \{y_0, \dots, y_n\} \quad (x_k = \phi(y_k))
 \end{aligned}$$

Under this 1-1 correspondance, we have

$$\begin{aligned}
 \forall 1 \leq k \leq n \quad & \phi([y_{k-1}, y_k]) = [x_{k-1}, x_k] \\
 \forall 1 \leq k \leq n \quad & \Delta\beta_k = \beta(y_k) - \beta(y_{k-1}) = \alpha(\phi(y_k)) - \alpha(\phi(y_{k-1})) = \alpha(x_k) - \alpha(x_{k-1}) = \Delta\alpha_k \\
 \forall 1 \leq k \leq n \quad & M_k^{(g)} = \sup \{g(y) : y \in [y_{k-1}, y_k]\} = \sup \{f(x) : x \in [x_{k-1}, x_k]\} = M_k^{(f)} \\
 \forall 1 \leq k \leq n \quad & m_k^{(g)} = \inf \{g(y) : y \in [y_{k-1}, y_k]\} = \inf \{f(x) : x \in [x_{k-1}, x_k]\} = m_k^{(f)}.
 \end{aligned}$$

Thus, under the above correspondance,

$$\begin{aligned}
 U(f, \alpha, P) &= U(g, \beta, Q) \\
 L(f, \alpha, P) &= L(g, \beta, Q)
 \end{aligned}$$

In order to show $g \in R_\beta[A, B]$, by the Cauchy Criterion, it is enough to prove that

$$\forall \epsilon > 0 \quad \exists Q \in \Pi[A, B] \text{ such that } U(g, \beta, Q) - L(g, \beta, Q) < \epsilon.$$

Let $\epsilon > 0$ be given. Since $f \in R_\alpha[a, b]$, $\exists P \in \Pi[a, b]$ such that

$$U(f, \alpha, P) - L(f, \alpha, P) < \epsilon.$$

We claim that $Q = H(P)$ is the partition that we're looking for. Indeed,

$$U(g, \beta, H(P)) - L(g, \beta, H(P)) = U(f, \alpha, P) - L(f, \alpha, P) < \epsilon.$$

Also,

$$\begin{aligned}\int_a^b f d\alpha &= L(f, \alpha) = \sup \{L(f, \alpha, P) : P \in \Pi[a, b]\} \\ &= \sup \{L(g, \beta, H(P)) : H(P) \in \Pi[A, B]\} \\ &= L(g, \beta) \\ &= \int_A^B g d\beta.\end{aligned}$$

□

Theorem 2.4.3. (The Fundamental Theorem of Calculus)

(i) Suppose $f \in R[a, b]$ and $F : [a, b] \rightarrow \mathbb{R}$ satisfies $F'(x) = f(x) \quad \forall x \in [a, b]$. Then

$$\int_a^b f = F(b) - F(a).$$

(ii) Suppose $g \in R[a, b]$ and $G : [a, b] \rightarrow \mathbb{R}$ is defined by $G(x) = \int_a^x g(t)dt$. Then

- (1) G is uniformly continuous on $[a, b]$.
- (2) If g is differentiable at $c \in [a, b]$, then G is differentiable at c and

$$G'(c) = g(c).$$

Proof. (i) In what follows, we'll show that

$$\forall P \in \Pi[a, b] \quad L(f, P) \leq F(b) - F(a) \leq U(f, P). \quad (*)$$

As a consequence of this:

(I) $F(b) - F(a)$ is an upper bound for $\{L(f, P) : P \in \Pi\}$, so

$$\sup \{L(f, P) : P \in \Pi\} \leq F(b) - F(a) \implies L(f) \leq F(b) - F(a)$$

(II) $F(b) - F(a)$ is a lower bound for $\{U(f, P) : P \in \Pi\}$, so

$$\inf \{U(f, P) : P \in \Pi\} \geq F(b) - F(a) \implies U(f) \geq F(b) - F(a).$$

Thus, $L(f) \leq F(b) - F(a) \leq U(f)$. Since $f \in R[a, b]$, $U(f) = L(f) = \int_a^b f d\alpha$, giving us

$$\int_a^b f d\alpha \leq F(b) - F(a) \leq \int_a^b f d\alpha.$$

Therefore, $\int_a^b f d\alpha$.

So, we just need to prove (*). Let $P = \{x_0, \dots, x_n\} \in \Pi[a, b]$. We have

$$\begin{aligned}F(b) - F(a) &= \sum_{k=1}^n F(x_k) - F(x_{k-1}) \\ &= \sum_{k=1}^n F'(t_k)(x_k - x_{k-1}) \text{ for some } t_k \in [x_{k-1}, x_k] \quad (\text{MVT for derivatives}) \\ &= \sum_{k=1}^n f(t_k) \Delta x_k.\end{aligned}$$

Therefore, it follows from

$$L(f, p) = \sum_{k=1}^n m_k \Delta x_k \leq \sum_{k=1}^n f(t_k) \Delta x_k \leq \sum_{k=1}^n M_k \Delta x_k = U(f, P)$$

that

$$L(f, P) \leq F(b) - F(a) \leq U(f, P).$$

- (ii) (1) Our goal is to show G is uniformly continuous on $[a, b]$. That is, we need to show

$$\forall \epsilon > 0 \exists \delta > 0 \text{ such that } \forall x, y \in [a, b] \text{ if } |x - y| < \delta, \text{ then } |G(x) - G(y)| < \epsilon.$$

Let $\epsilon > 0$ be given. Our goal is to find $\delta > 0$ such that

$$\forall x, y \in [a, b] \text{ if } |x - y| < \delta, \text{ then } |G(x) - G(y)| < \epsilon. \quad (*)$$

Let $x, y \in [a, b]$. If $x \geq y$,

$$\begin{aligned} |G(x) - G(y)| &= \left| \int_a^x g(t) dt - \int_a^y g(t) dt \right| \\ &= \left| \int_y^x g(t) dt \right| \\ &\leq \int_y^x |g(t)| dt \\ &\leq \int_y^x R dt \quad (\text{Let } R = \sup_{a \leq t \leq b} |g(t)|) \\ &= R(x - y) \\ &= R|x - y|. \end{aligned}$$

If $x < y$, then

$$\begin{aligned} |G(x) - G(y)| &= |G(y) - G(x)| \\ &= \left| \int_a^y g(t) dt - \int_a^x g(t) dt \right| \\ &= \left| \int_x^y g(t) dt \right| \\ &\leq \int_x^y |g(t)| dt \\ &\leq \int_x^y R dt \\ &= R(y - x) \\ &= R|x - y|. \end{aligned}$$

Thus, $\forall x, y \in [a, b]$

$$|G(x) - G(y)| \leq R|x - y|.$$

Hence, to ensure $|G(x) - G(y)| < \epsilon$ it is enough to make $R|x - y| < \epsilon$. That is, it is enough to make $|x - y| < \epsilon/R$. Choosing $\delta = \epsilon/R$ does the job, and so $(*)$ is satisfied.

- (2) Suppose g is continuous at $c \in [a, b]$. Our goal is to show $G'(c) = g(c)$. That is, we want to show

$$\lim_{x \rightarrow c} \frac{G(x) - G(c)}{x - c} = g(c).$$

That is, our goal is to show

$$\forall \epsilon > 0 \exists \delta > 0 \text{ such that if } 0 < |x - c| < \delta, \text{ then } \left| \frac{G(x) - G(c)}{x - c} - g(c) \right| < \epsilon.$$

Let $\epsilon > 0$ be given. Since g is continuous at c , for the given ϵ , there exists $\hat{\delta} > 0$ such that

$$\text{if } |t - c| < \hat{\delta}, \text{ then } |g(t) - g(c)| < \epsilon.$$

$t \in [a, b]$

We claim that this $\hat{\delta}$ can be used as the δ we're looking for. Indeed, if we let $\delta = \hat{\delta}$, then

Case 1: $0 < |x - c| < \hat{\delta}, x \in [a, b], x > c$

$$\begin{aligned}
 \left| \frac{G(x) - G(c)}{x - c} - g(c) \right| &= \left| \frac{G(x) - G(c) - g(c)(x - c)}{x - c} \right| \\
 &= \left| \frac{\int_a^x g(t)dt - \int_a^c g(t)dt - \int_a^x g(c)dt}{x - c} \right| \\
 &= \left| \frac{\int_c^x g(t)dt - \int_c^x g(c)dt}{x - c} \right| \\
 &= \left| \frac{\int_c^x g(t) - g(c)dt}{x - c} \right| \\
 &= \frac{1}{|x - c|} \left| \int_c^x g(t) - g(c)dt \right| \\
 &\leq \frac{1}{|x - c|} \int_c^x |g(t) - g(c)|dt \\
 &\leq \frac{1}{x - c} \int_c^x \frac{\epsilon}{2} dt \\
 &= \frac{\epsilon/2}{x - c} (x - c) < \epsilon.
 \end{aligned}$$

Case 2: $0 < |x - c| < \hat{\delta}, x \in [a, b], x < c$

$$\begin{aligned}
 \left| \frac{G(x) - G(y)}{x - c} - g(c) \right| &= \left| \frac{G(x) - G(y) - g(c)(x - c)}{x - c} \right| \\
 &= \left| \frac{g(c)(c - x) - (G(c) - G(x))}{c - x} \right| \\
 &= \left| \frac{\int_x^c g(c)dt - \int_a^c g(t)dt + \int_a^x g(t)dt}{c - x} \right| \\
 &\vdots \\
 &= \left| \frac{\int_x^c g(c) - g(t)dt}{c - x} \right| \\
 &\leq \frac{1}{c - x} \int_x^c |g(c) - g(t)|dt \\
 &\leq \frac{\int_x^c \frac{\epsilon}{2} dt}{c - x} \\
 &< \epsilon.
 \end{aligned}$$

□

Notation . Let I be a nonempty interval in $\mathbb{R}$.

$$C^0(I) = C(I) = \{f : I \rightarrow \mathbb{R} \mid f \text{ is continuous}\}$$

$$\forall n \geq 1 \quad C^n(I) = \left\{ f : I \rightarrow \mathbb{R} \mid f, f', \dots, f^{(n)} \text{ exist and are continuous} \right\}$$

Theorem 2.4.4. (Integration by Parts)

Suppose $u : [a, b] \rightarrow \mathbb{R}$ and $v : [a, b] \rightarrow \mathbb{R}$ are differentiable and $u', v' \in R[a, b]$. Then

(i) $uv' \in R[a, b]$.

(ii) $u'v \in R[a, b]$.

(iii) $\int_a^b uv'dx = u(b)v(b) - u(a)v(a) - \int_a^b u'vdx$.

Proof. (i) $u : [a, b] \rightarrow \mathbb{R}$ is differentiable, so

$$\begin{aligned} u \in C[a, b] &\implies u \in R[a, b] \\ \left. \begin{aligned} u \in R[a, b] \\ v' \in R[a, b] \end{aligned} \right\} &\implies uv' \in R[a, b] \end{aligned}$$

(ii) Analogous to (i).

(iii) By the product rule,

$$(uv)' = u'v + uv'.$$

In particular, since $(uv)'$ is the sum of integrable functions it follows that $(uv)' \in R[a, b]$. Integrate both sides:

$$\int_a^b (uv)' dx = \int_a^b u'v dx = \int_a^b uv' dx \quad (I)$$

By the Fundamental Theorem of Calculus part (i),

$$\int_a^b (uv)' dx = [uv]_{x=a}^{x=b} = u(b)v(b) - u(a)v(a) \quad (II)$$

By (I) and (II), we have

$$\begin{aligned} u(b)v(b) - u(a)v(a) &= \int_a^b u'v dx + \int_a^b uv' dx \\ \implies \int_a^b u'v dx &= u(b)v(b) - u(a)v(a) - \int_a^b uv' dx \end{aligned}$$

□

Definition 2.4.1. (Unit Step Function)

The unit step function $I : \mathbb{R} \rightarrow \mathbb{R}$ is defined by

$$I(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ 1 & \text{if } x > 0 \end{cases}.$$

Note that for all $s \in \mathbb{R}$,

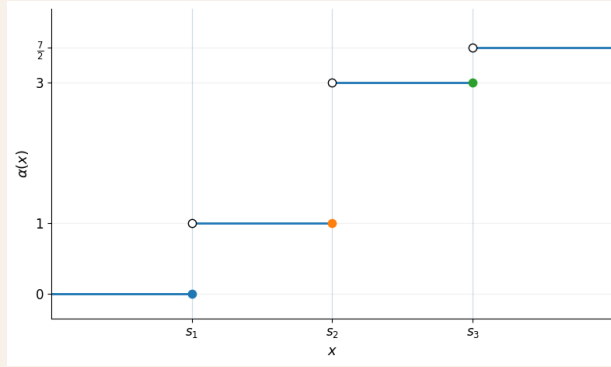
$$I(x - s) = \begin{cases} 0 & \text{if } x \leq s \\ 1 & \text{if } x > s \end{cases}.$$

Also, for all $c \neq 0$,

$$cI(x - s) = \begin{cases} 0 & \text{if } x \leq s \\ 1 & \text{if } x > s \end{cases}.$$

Example 2.4.1. Suppose $s_1 < s_2 < s_3$. Graph the function

$$\begin{aligned} \alpha(x) &= I(x - s_1) + 2I(x - s_2) + \frac{1}{2}I(x - s_3) \\ \implies \alpha(x) &= \begin{cases} 0 + 0 + 0 & \text{if } x \leq s_1 \\ 1 + 0 + 0 & \text{if } s_1 < x \leq s_2 \\ 1 + 2 + 0 & \text{if } s_2 < x \leq s_3 \\ 1 + 2 + \frac{1}{2} & \text{if } s_3 < x \end{cases} \end{aligned}$$

**Theorem 2.4.5.**

Suppose $f : [a, b] \rightarrow \mathbb{R}$ is bounded, $s \in (a, b)$, f is right continuous at s , and $\alpha(x) = I(x - s)$. Then $f \in R_\alpha[a, b]$ and $\int_a^b f d\alpha = f(s)$.

Proof. In what follows, we will introduce a sequence of partitions (P_n) in $\Pi[a, b]$ such that

$$\lim_{n \rightarrow \infty} [U(f, \alpha, P_n) - L(f, \alpha, P_n)] = 0 \text{ and } \lim_{n \rightarrow \infty} U(f, \alpha, P_n) = \lim_{n \rightarrow \infty} L(f, \alpha, P_n) = f(s).$$

This, according to the sequential criterion of integrability, tells us that

$$f \in R_\alpha[a, b] \text{ and } \int_a^b f d\alpha = f(s).$$

Let N be large enough so that $s + \frac{1}{N} < b$. For each $n \geq N$, let

$$P_n = \left\{ a, s, s + \frac{1}{n}, b \right\}.$$

We have

$$\begin{aligned} U(f, \alpha, P_n) &= \sum_{k=1}^3 M_k \Delta \alpha_k = M_2 \Delta \alpha_2 = \left[\sup_{s \leq x \leq s + \frac{1}{n}} f(x) \right] (1) = \sup_{s \leq x \leq s + \frac{1}{n}} f(x) \\ L(f, \alpha, P_n) &= \sum_{k=1}^3 m_k \Delta \alpha_k = m_2 \Delta \alpha_2 = \left[\inf_{s \leq x \leq s + \frac{1}{n}} f(x) \right] (1) = \inf_{s \leq x \leq s + \frac{1}{n}} f(x). \end{aligned}$$

Therefore,

$$\begin{aligned} \lim_{n \rightarrow \infty} U(f, \alpha, P_n) &= \lim_{n \rightarrow \infty} \sup_{s \leq x \leq s + \frac{1}{n}} f(x) = f(s) \\ \lim_{n \rightarrow \infty} L(f, \alpha, P_n) &= \lim_{n \rightarrow \infty} \inf_{s \leq x \leq s + \frac{1}{n}} f(x) = f(s). \end{aligned}$$

Here, we are using the fact that f is continuous at s , but right continuity would have been enough. We conclude that $\lim_{n \rightarrow \infty} [U(f, \alpha, P_n) - L(f, \alpha, P_n)] = f(s) - f(s) = 0$, so $f \in R_\alpha[a, b]$ and $\int_a^b f d\alpha = \lim_{n \rightarrow \infty} U(f, \alpha, P_n) = f(s)$. $\square$

Theorem 2.4.6.

Suppose $\forall n \geq 1, c_n \geq 0$, $\sum_{n=1}^{\infty} c_n$ converges. Let $s_1 < s_2 < \dots$ be points in (a, b) , $\alpha : [a, b] \rightarrow \mathbb{R}$, $\alpha(x) = \sum_{n=1}^{\infty} c_n I(x - s_n)$, and $f : [a, b] \rightarrow \mathbb{R}$ be continuous. Then

$$f \in R_\alpha[a, b] \text{ and } \int_a^b f d\alpha = \sum_{n=1}^{\infty} c_n f(s_n).$$

Chapter 3

Sequences of Functions

3.1 Sequences of Functions and Pointwise Convergence

Definition 3.1.1. (Sequences of Functions)

Let $A \neq \emptyset$. If $\forall n \in \mathbb{N}$ we have $f_n : A \rightarrow \mathbb{R}$, then

$$f_1, f_2, f_3, \dots$$

is a sequence of functions and is denoted

$$(f_n)_{n \geq 1}, (f_n), \text{ or } (f_n : A \rightarrow \mathbb{R})_{n \geq 1}.$$

Example 3.1.1. (i) $(f_n)_{n \geq 1} = \left(\frac{\sin(nx)}{n} \right)_{n \geq 1}$ on the set $\mathbb{R}$:

$$f_1 : \mathbb{R} \rightarrow \mathbb{R}, f_1(x) = \sin x$$

$$f_2 : \mathbb{R} \rightarrow \mathbb{R}, f_2(x) = \frac{1}{2} \sin 2x$$

$$f_3 : \mathbb{R} \rightarrow \mathbb{R}, f_3(x) = \frac{1}{3} \sin 3x$$

$\vdots$

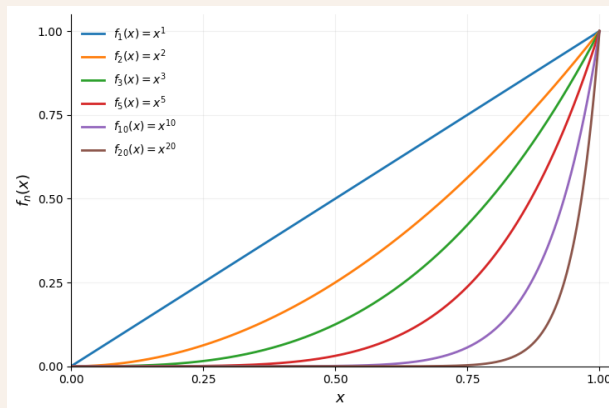
(ii) $(f_n)_{n \geq 1} = (x^n)$ on the set $[0, 1]$:

$$f_1 : [0, 1] \rightarrow \mathbb{R}, f_1(x) = x$$

$$f_2 : [0, 1] \rightarrow \mathbb{R}, f_2(x) = x^2$$

$$f_3 : [0, 1] \rightarrow \mathbb{R}, f_3(x) = x^3$$

$\vdots$



(iii) Let $f : [0, 1] \rightarrow \mathbb{R}$ be a given function. For all $n \in \mathbb{N}$, let $f_n(x) = f(x^n)$. Then (f_n) is a sequence of functions.

Remark 3.1.1. Suppose $(f_n : A \rightarrow \mathbb{R})_{n \geq 1}$ is a sequence of functions. Note that for each point $c \in A$, $(f_n(c))_{n \geq 1}$ will be a sequence of numbers. For example, consider $(f_n(x) = x^n)_{n \geq 1}$ on the set $A = [0, 2]$:

$x = 0$: $f_1(0) = 0, f_2(0) = 0, f_3(0) = 0, \dots$ is a sequence of numbers

$x = 1$: $f_1\left(\frac{1}{2}\right) = \frac{1}{2}, f_2\left(\frac{1}{2}\right) = \frac{1}{2^2}, f_3\left(\frac{1}{2}\right) = \frac{1}{2^3}, \dots$ is a sequence of numbers

$x = 2$: $f_1(1) = 1, f_2(1) = 1, f_3(1) = 1, \dots$ is a sequence of numbers

$\vdots$

Definition 3.1.2. (Pointwise Convergent Sequence)

We say a sequence of functions $(f_n : A \rightarrow \mathbb{R})_{n \geq 1}$ is pointwise convergent if at any $c \in A$, the sequence $(f_n(c))$ converges.

Example 3.1.2. For all $n \geq 1$, $f_n : \{0, 1\} \rightarrow \mathbb{R}, f_n(x) = 2x + \frac{x}{n}$.

$c \in \{0, 1\}$:	$(f_n(c))$	Convergence
0	$(f_n(0)) = (0, 0, 0, \dots)$	$f_n(0) \rightarrow 0$
1	$(f_n(1)) = (2 + 1, 2 + \frac{1}{2}, 2 + \frac{1}{3}, \dots)$	$f_n(1) \rightarrow 2$

So, (f_n) is a pointwise convergent sequence.

Definition 3.1.3. (Pointwise Limit)

Let $(f_n : A \rightarrow \mathbb{R})_{n \geq 1}$ be a pointwise convergent sequence. Then we define a new function on A as follows:

$$f : A \rightarrow \mathbb{R}, \quad \forall c \in A \quad f(c) = \lim_{n \rightarrow \infty} f_n(c).$$

We call f the pointwise limit of the sequence (f_n) on A . We say f_n converges pointwise to the function f on A .

Example 3.1.3. For all $n \geq 1$, $f_n : [-1, 1] \rightarrow \mathbb{R}, f_n(x) = x^2 + \frac{1}{n}$. Prove that $f_n \rightarrow f$ pointwise on $[-1, 1]$ where $f(x) = x^2$.

At each point $c \in [-1, 1]$, the corresponding sequence of numbers

$$(f_n(c))_{n \geq 1} = (c^2 + 1, c^2 + \frac{1}{2}, c^2 + \frac{1}{3}, \dots) = (c^2 + \frac{1}{n})_{n \geq 1}.$$

For each $c \in [-1, 1]$, we have

$$\lim_{n \rightarrow \infty} f_n(c) = \lim_{n \rightarrow \infty} \left(c^2 + \frac{1}{n} \right) = c^2 = f(c).$$

Thus, $f_n \rightarrow f$ pointwise where $f : [-1, 1] \rightarrow \mathbb{R}$ defined by $f(x) = x^2$.

Example 3.1.4. For all $n \geq 1$, $f_n : [0, 1] \rightarrow \mathbb{R}, f_n(x) = \frac{x}{n}$. Prove that $f_n \rightarrow f$ where $f : [0, 1] \rightarrow \mathbb{R}$ is defined by $f(x) \equiv 0$.

At each point $c \in [0, 1]$, the corresponding sequence of numbers is

$$(f_n(c)) = \left(c, \frac{c}{2}, \frac{c}{3}, \dots \right) = \left(\frac{c}{n} \right)_{n \geq 1}.$$

Thus for each $c \in [0, 1]$, we have

$$\lim_{n \rightarrow \infty} f_n(c) = \lim_{n \rightarrow \infty} \frac{c}{n} = c \lim_{n \rightarrow \infty} \frac{1}{n} = c \cdot 0 = 0.$$

Example 3.1.5. $\forall n \geq 1$ $f_n : [0, 1] \rightarrow \mathbb{R}, f_n(x) = x^n$. Prove that $f_n \rightarrow f$ pointwise where $f : [0, 1] \rightarrow \mathbb{R}$ is defined by $f(x) = \begin{cases} 0 & \text{if } 0 \leq x < 1 \\ 1 & \text{if } x = 1 \end{cases}$.

At each point $c \in [0, 1]$, the corresponding sequence of numbers is

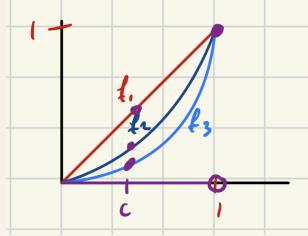
$$(f_c(c))_{n \geq 1} = (c^n)_{n \geq 1} = (c, c^2, c^3, \dots).$$

Note that

$$(*) \text{ if } 0 \leq c < 1, \text{ then } f_n(c) \rightarrow c^n = 0 = f(c).$$

$$(*) \text{ if } c = 1, \text{ then } f_n(c) \rightarrow 1 = f(c).$$

So, $f_n \rightarrow f$ pointwise.



Observation 3. In this example, each f_n is continuous but the pointwise limit f is not continuous.

Remark 3.1.2. The pointwise limit of a sequence of continuous functions is not necessarily continuous.

”Pointwise convergence does not preserve convergence.”

Example 3.1.6. $\forall n \geq 1 \ f_n : [0, 1] \rightarrow \mathbb{R}, f_n(x) = \begin{cases} x & \text{if } x \leq n \\ n & \text{if } x > n \end{cases}$. Prove $f_n \rightarrow f$ pointwise where $f : [0, \infty) \rightarrow \mathbb{R}$ is defined by $f(x) = x$.

At each point $c \in [0, \infty)$, the corresponding sequence of numbers is

$$(f_n(c))_{n \geq 1} = (f_1(c), f_2(c), f_3(c), \dots).$$

Note that for fixed $c \in [0, \infty)$, $\exists N \in \mathbb{N}$ such that $c \leq N$, so clearly

$$c \leq N, \quad c \leq N + 1, \quad c \leq N + 2, \dots$$

Thus

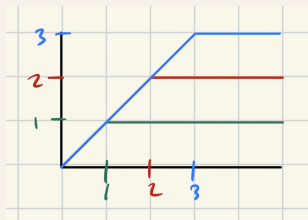
$$f_N(c) = c, f_{N+1}(c) = c, f_{N+2}(c) = c, \dots$$

Therefore,

$$(f_n(c))_{n \geq 1} = (f_1(c), f_2(c), f_3(c), \dots, f_{N-1}(c), c, c, c, \dots).$$

We conclude that

$$\lim_{n \rightarrow \infty} f_n(c) = c.$$



Observation 4. Each f_n is bounded, but $f(x) = x$ is not bounded.

Remark 3.1.3. The pointwise limit of a sequence of bounded functions is not necessarily bounded.

”Pointwise convergence does not preserve boundedness.”

Example 3.1.7. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous, nowhere differentiable function. By the Weierstrass Approximation Theorem, there exists a sequence of polynomials $p_1(x), p_2(x), p_3(x), \dots$ such that p_n converges to f pointwise.

Observation 5. Each p_n is differentiable, but the pointwise limit f is not differentiable.

Remark 3.1.4. The pointwise limit f of a sequence of differentiable functions is not necessarily differentiable. Also, if f' exists, it's not the case that $f'_n \rightarrow f'$.

"Pointwise convergence does not preserve differentiability."

Example 3.1.8. $\forall n \geq 4$, define $f_n : [0, 1] \rightarrow \mathbb{R}$ as follows:

$$f_n(x) = \begin{cases} 0 & \text{if } 0 \leq x < \frac{1}{n} \\ n(nx - 1) & \text{if } \frac{1}{n} \leq x < \frac{2}{n} \\ -n(nx - 3) & \text{if } \frac{2}{n} \leq x < \frac{3}{n} \\ 0 & \text{if } \frac{3}{n} \leq x \leq 1 \end{cases}.$$

Prove that $f_n \rightarrow f$ pointwise where $f \equiv 0$.

At each fixed point $c \in [0, 1]$, the corresponding sequence of numbers

$$(f_n(c))_{n \geq 4} = (f_4(c), f_5(c), f_6(c), \dots).$$

Note that

$$\begin{aligned} c = 0 : (f_n(0))_{n \geq 4} &= (0, 0, 0, 0, \dots) \implies \lim_{n \rightarrow \infty} f_n(0) = 0 \\ 0 < c \leq 1 : \exists N \in \mathbb{N} \text{ such that } \frac{3}{N} < c &\implies \frac{3}{N} < c, \frac{3}{N+1} < c, \frac{3}{N+2} < c, \dots \\ &\implies f_N(c) = 0, f_{N+1}(c) = 0, f_{N+2}(c) = 0, \dots \\ &\implies (f_n(c))_{n \geq 4} = (f_4(c), f_5(c), \dots, f_{N-1}(c), 0, 0, \dots) \\ &\implies \lim_{n \rightarrow \infty} f_n(c) = 0. \end{aligned}$$

$\forall c \in [0, 1], \lim_{n \rightarrow \infty} f_n(c) = 0 = f(c)$, so $f_n \rightarrow f = 0$ pointwise.

Observation 6. Each $f_n, \int_0^n f_n = 1$. However, $\int_0^1 f = 0$, so for every $n, \int_0^1 f_n \neq \int_0^1 f$.

Remark 3.1.5. Pointwise convergence does not allow us to interchange limits and integrals.

"Pointwise convergence does not preserve integrals."